KOMOREBI MATHEMATICS / PAPER III

Projector-Resolved Seed Calculus III

Typed Transfer, Nonlinear Packets, and Obstruction Theory

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Reading route. Typed edge words and exact coordinate transport lead to connection defects and packet obstruction witnesses. Nonlinear packet jets then lead to tree expansions, correction complexes, filtered lifting, elimination, and smooth transport.

Scope. Part I retains the finite v1.0.1 formal contract. Part II gives the strengthened nonlinear and analytic development with ordinary proofs. Formal coverage stays attached to the exact released declarations and does not extend merely because both parts appear in one edition.

Abstract

Finite projector packets turn linear transfer into typed walks and arbitrary maps into exact coordinate representations. We separate canonical coordinate transport, which satisfies an exact cocycle law, from a chosen connection, whose failure to compose produces holonomy and packet obstruction witnesses. We then develop nonlinear packet jets and higher composition, convergent rooted tree expansions, graded operations, and nonlinear refinement defects. A specified correction map produces a canonical cokernel obstruction. Relative exact sequences, mapping cones, Hilbert splitting, nonlinear correction, and filtered lifting describe how these obstructions behave. Exact elimination produces Schur formulas, and finite transport reduces gluing to fundamental cycle holonomy. For smooth orthogonal packet families, direct packet rotations converge in operator norm to canonical unitary transport. The finite formal contract and the broader ordinary proof development are identified separately throughout.

This edition preserves both the finite release and the strengthened nonlinear manuscript. Its companions are a worked visual guide and a Lean guide. The released Lean evidence is inherited; this editorial build does not assert a new Lean execution.

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Part I

Finite typed transfer and the formal contract

The finite release supplies the algebraic foundation and its declaration-level contract. In this part, "roadmap" and "outside the contract" refer to the finite Lean formalization, not to the absence of ordinary mathematical results in Part II. The original thirty theorem-like statements and their proof bodies are retained. The source contract has 41 mapping rows, which include definitions and other interfaces and therefore are not the same count as theorem-like statements.

The preserved v1.0.1 release records Lean v4.34.0-rc2 against Mathlib revision `aeb72f1e9981b6d3 5b426b16dca7759be3e7ba06` [3, 17]. The compiler and kernel-check descriptions in Part I describe that inherited evidence. Current editorial checks of this edition are recorded separately in the companion verification files.

1 Purpose and status discipline

Paper I fixed typed seeds, operator grammars, exact projector packets, generated hulls, and reconstruction [13]. Paper II added coherent refinement coordinates, weighted transfer, arithmetic separation, and linear defect identities [14]. The present paper begins the next layer: explicit edge words, genuinely optional connection data, and packet-local obstructions.

The central distinction is categorical rather than terminological. Canonical coordinate transport compares two exact descriptions of the *same* ambient object and is forced by analysis and synthesis. A connection chooses transports that may carry extra modeling, discretization, or gauge information. The former obeys a cocycle theorem; failure of the latter to obey a cocycle is meaningful data. Treating both maps as one symbol would make holonomy either tautologically trivial or logically unfounded.

1.1 Statement labels

Every substantial theorem, lemma, corollary, proposition, definition/API, and roadmap item carries status evidence drawn from one or more of three labels. A statement whose separate clauses use both a Paper III result and an inherited bridge carries both applicable labels. Illustrative examples are exempt.

S1. Lean-backed in Paper III: the displayed scope is no stronger than the named compiled declaration in the released Paper III module.

S2. Inherited Lean-backed: the displayed result is supplied by the frozen formal contract of Paper I or II.

S3. Roadmap only: the text states a future interface or theorem target and is excluded from the completion denominator.

Boundary 1.1 (Final theorem denominator). The completion denominator of the finite v1.0.1 contract is the thirty theorem-like environments in Part I: twenty-six resolve to compiled Paper III declarations and four resolve to the frozen Papers I–II contracts. The authoritative theorem map names the exact declarations used by each statement. The kernel audit checks those targets and

supporting theorems, while a fail-closed gate scans the complete compiled `Komorebi.PaperIII` namespace. Roadmap items are excluded from the denominator and remain visibly marked.

2 Finite typed packet graphs

2.1 Orientation and matrix convention

Let V be a finite packet type, E a finite edge type, and R a semiring. A directed multigraph is a pair of endpoint functions

$$\mathrm{src}, \mathrm{tgt} : E \longrightarrow V.$$

Parallel edges remain distinct elements of E. Rows are targets and columns are sources. Thus an edge $e : i \to j$ is represented by a matrix supported at row j, column i.

Definition 2.1 (Packet graph and scalar edge system)**.** For $a, b \in V$ and $r \in R$, define

$$\mathrm{packetMatrix}(a, b; r)_{yx} = \begin{cases} r, & y = b \text{ and } x = a, \\ 0, & \text{otherwise.} \end{cases}$$

An edge system is a coefficient function $c : E \to R$. Its elementary edge operator and total transfer operator are

$$L_e = \mathrm{packetMatrix}(\mathrm{src}(e), \mathrm{tgt}(e); c(e)), \qquad T = \sum_{e \in E} L_e.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph;Komorebi.PaperIII.EdgeOperators.PacketGraph.packetMatrix;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp`

The convention is execution ordered. In $L_f L_e$, the rightmost operator L_e acts first. For a written word $(e_1, \ldots, e_n)$, e_1 acts first and the operator product is $L_{e_n} \cdots L_{e_1}$.

Lemma 2.2 (Declared support)**.** *For every edge e, $(L_e)_{yx} = 0$ unless $y = \mathrm{tgt}(e)$ and $x = \mathrm{src}(e)$.*

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_eq_zero_off_support`

Theorem 2.3 (Typed interface annihilation)**.** *If $\mathrm{tgt}(e) \neq \mathrm{src}(f)$, then*

$$L_f L_e = 0.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_mul_edgeOp_eq_zero`

Proof. For a matrix entry (y, x),

$$(L_f L_e)_{yx} = \sum_{z \in V} (L_f)_{yz} (L_e)_{zx}.$$

A potentially nonzero summand would require simultaneously $z = \mathrm{src}(f)$ and $z = \mathrm{tgt}(e)$, contrary to the hypothesis. □

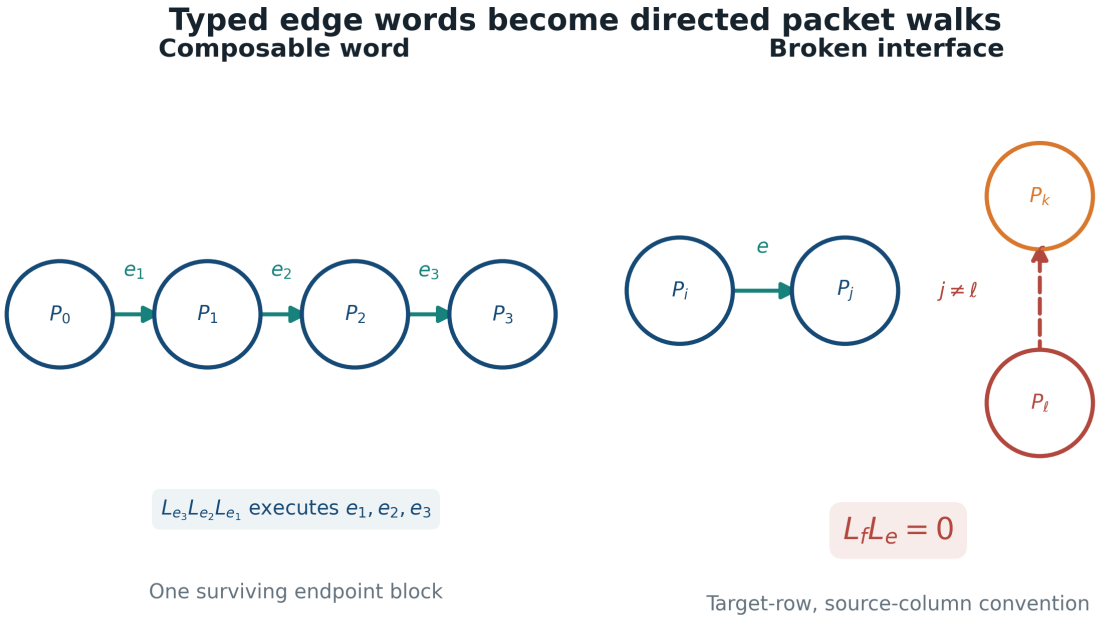


Figure 1. The target-row, source-column convention fixes execution order. Composable words collapse to endpoint blocks, while one broken packet interface annihilates the entire operator product.

Theorem 2.4 (Matched two-edge collapse)**.** *If* $\mathrm{tgt}(e) = \mathrm{src}(f)$*, then*

$$L_f L_e = \mathrm{packetMatrix}\big(\mathrm{src}(e), \mathrm{tgt}(f); c(f)c(e)\big).$$

Together with theorem 2.3, this gives the complete composable/noncomposable dichotomy for two edge operators.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.edgeOp_mul_edgeOp_of_composable`

Proof. The matrix-product sum has exactly one potentially nonzero intermediate index, namely $\mathrm{tgt}(e) = \mathrm{src}(f)$. At that index the coefficients occur in operator order as $c(f)c(e)$, and the outer source and target are $\mathrm{src}(e)$ and $\mathrm{tgt}(f)$. □

Boundary 2.5 (Current fiber scope). The compiled Paper III model above has one scalar coordinate per packet. For a family of modules $(X_i)_{i \in V}$, an intrinsic edge map would instead have type $\ell_e : X_{\mathrm{src}(e)} \to X_{\mathrm{tgt}(e)}$, and its totalization on $\bigoplus_i X_i$ would be $\iota_{\mathrm{tgt}(e)} \ell_e \pi_{\mathrm{src}(e)}$. The same support proof applies, but this dependent fiber-valued generalization is not attributed to the released scalar Paper III edge-system modules.

Figure 1 summarizes the support convention and the composable/noncomposable word dichotomy used throughout the next section.

3 Operator words and directed walks

3.1 Typed words

Definition 3.1 (Composable word and execution operator)**.** A finite word $w = (e_1, \dots, e_n)$ is composable when

$$\mathrm{tgt}(e_r) = \mathrm{src}(e_{r+1}) \qquad (1 \le r < n).$$

Set

$$L_w = L_{e_n} \cdots L_{e_1}, \qquad L_\varnothing = I.$$

This definition records the written order of execution even though matrix composition is read right to left.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Composable;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp`

Theorem 3.2 (First broken interface)**.** *If a word begins* $(e, f, \dots)$ *and* $\mathrm{tgt}(e) \neq \mathrm{src}(f)$*, then its word operator is zero.*

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_zero_of_first_break`

Theorem 3.3 (Nonwalk words vanish)**.** *If* w *is not composable, then* $L_w = 0$*.*

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_zero_of_not_composable`

Proof. Choose the first broken adjacent pair. The corresponding consecutive factor is zero by theorem 2.3, and associativity absorbs that zero into the complete product. The Lean proof implements the same argument by induction on the list. □

3.2 The checked recursive expansion

Definition 3.4 (Recursive word expansion)**.** Define

$$\mathcal{W}_0 = I, \qquad \mathcal{W}_{n+1} = T\mathcal{W}_n.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordExpansion`

Theorem 3.5 (Transfer power equals recursive word expansion)**.** *For every* $n \in \mathbb{N}$*,*

$$T^n = \mathcal{W}_n.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_wordExpansion`

Proof. The zero case is the identity matrix. If $T^n = \mathcal{W}_n$, then the left-successor power convention gives $T^{n+1} = TT^n = T\mathcal{W}_n = \mathcal{W}_{n+1}$. □

Remark 3.6. The formal module also defines an explicit sum over tuples $w : \mathrm{Fin}(n) \to E$ and a second sum in which noncomposable tuples are replaced by zero. It proves that the recursive, explicit, and walk-filtered matrices agree. These are separate declarations so that the finite enumeration claim is auditable rather than hidden in the name *word expansion*.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.walkExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordExpansion_eq_explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.explicitWordExpansion_eq_walkExpansion`

3.3 Exact word-to-walk expansion

For a nonempty composable word $w = (e_1, \dots, e_n)$, define its reverse coefficient product

$$c^{\leftarrow}(w) = c(e_n)c(e_{n-1}) \cdots c(e_1).$$

The reversal is essential over a noncommutative semiring.

Lemma 3.7 (One word is one endpoint block)**.** *If $w = (e_1, \dots, e_n)$ is nonempty and composable, then*

$$L_w = \mathrm{packetMatrix}\big(\mathrm{src}(e_1), \mathrm{tgt}(e_n); c^{\leftarrow}(w)\big).$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.reverseCoefficient;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.terminalTarget;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.wordOp_eq_packetMatrix_of_composable`

Proof. Induct on n. The case $n = 1$ is the definition of L_e. Suppose the formula holds for the prefix $(e_1, \dots, e_n)$ and append f. Composability gives $\mathrm{tgt}(e_n) = \mathrm{src}(f)$. Multiplying the two packet matrices leaves one intermediate index and changes the coefficient to $c(f)c(e_n) \cdots c(e_1)$, with source $\mathrm{src}(e_1)$ and target $\mathrm{tgt}(f)$. □

Write $\mathrm{Walk}_n(i, j)$ for the finite set of length-n composable edge tuples whose first source is i and whose final target is j. For $n = 0$, it has one empty walk when $i = j$ and is empty otherwise; assign the empty walk the coefficient 1.

Theorem 3.8 (Finite operator word-to-walk expansion)**.** *For every $n \in \mathbb{N}$,*

$$T^n = \sum_{w \in E^n} L_w = \sum_{w \in E^n} \begin{cases} L_w, & w \text{ composable}, \\ 0, & w \text{ not composable}. \end{cases}$$

Parallel edges occur as distinct summands.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_explicitWordExpansion;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_eq_walkExpansion`

Proof. Induction and finite distributivity give

$$\left(\sum_{e\in E} L_e\right)^n = \sum_{(e_1,\dots,e_n)\in E^n} L_{e_n}\cdots L_{e_1}.$$

Every noncomposable summand is zero by theorem 3.3, so inserting the composability filter does not change the sum. In Lean, tuple concatenation and the successor sum are handled by `tupleWordOp_snoc` and `explicitWordExpansion_succ`; the two bridge equalities then yield the displayed theorem. □

Corollary 3.9 (Endpoint coefficient walk sum)**.** *For all $n \in \mathbb{N}$ and $i, j \in V$,*

$$\boxed{(T^n)_{ji} = \sum_{w\in \mathrm{Walk}_n(i,j)} c^{\leftarrow}(w).}$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Connects;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.endpointWalkCoefficientSum;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_apply_eq_sum_walks`

Proof. Take the (j, i) entry in theorem 3.8. By lemma 3.7, a surviving word contributes $c^{\leftarrow}(w)$ exactly when its endpoints are i and j, and contributes zero otherwise. At $n = 0$, the identity matrix gives the unique empty diagonal walk. □

Corollary 3.10 (Fixed-length reachability obstruction)**.** *If* $\mathrm{Walk}_n(i, j) = \varnothing$*, then* $(T^n)_{ji} = 0$*.*

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_apply_eq_zero_of_no_walk`

Proof. This is immediate from the empty sum in the preceding corollary. □

Corollary 3.11 (Orbit support obstruction)**.** *Let $S \subseteq V$, and let $x \in R^V$ be supported on S. If j is not reachable by a directed path from any vertex of S, then*

$$(T^n x)_j = 0 \qquad \text{for every } n \in \mathbb{N}.$$

Hence the T-orbit of x cannot generate a vector with nonzero j-coordinate.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.Reachable;Komorebi.PaperIII.EdgeOperators.PacketGraph.EdgeSystem.transferOp_pow_mulVec_apply_eq_zero_of_unreachable_support`

Inherited formal contract: Papers I–II.
`Komorebi.PacketNaturalityCompletion.Reachability.algebraicOrbitHull_le_reachable_support;Komorebi.PacketNaturalityCompletion.Reachability.unreachable_nonzero_packet_obstruction`

Proof. Expand $(T^n x)_j = \sum_i (T^n)_{ji} x_i$. Terms with $i \notin S$ vanish by the support of x, and terms with $i \in S$ vanish by corollary 3.10. The final claim follows because the j-coordinate also vanishes on every finite linear combination of orbit vectors. □

4 Canonical exact coordinate transport

4.1 The Paper III frame model

Let every chart a carry a linear frame equivalence $F_a : M \xrightarrow{\sim} M$. The formal Paper III module defines transport from a to b by

$$J_{b\leftarrow a} = F_b^{-1} F_a.$$

This is the abstract frame form of *synthesize in chart a, then analyze in chart b.*

Theorem 4.1 (Exact frame transport is coherent)**.** *For every a, b, c,*

$$J_{a\leftarrow a} = \mathrm{id}, \qquad J_{a\leftarrow b} J_{b\leftarrow a} = \mathrm{id}, \qquad J_{c\leftarrow b} J_{b\leftarrow a} = J_{c\leftarrow a},$$

and therefore

$$J_{a\leftarrow c} J_{c\leftarrow b} J_{b\leftarrow a} = \mathrm{id}.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.exactTransport_self;Komorebi.PaperIII.Transport.exactTransport_roundTrip;Komorebi.PaperIII.Transport.exactTransport_cocycle;Komorebi.PaperIII.Transport.exactTransport_triangleLoop`

Proof. Every identity is cancellation of adjacent $F_r F_r^{-1}$ factors. The Lean definitions use `LinearEquiv.trans`; the displayed formulas use the equivalent right-to-left function-composition notation fixed in this paper. □

Theorem 4.2 (Every finite canonical path telescopes)**.** *Let $p = (a_0, a_1, \ldots, a_n)$ be any finite vertex list and put*

$$J_p = J_{a_n\leftarrow a_{n-1}} \cdots J_{a_1\leftarrow a_0}, \qquad J_{(a_0)} = \mathrm{id}.$$

Then

$$J_p = J_{a_n\leftarrow a_0}.$$

In particular, if $a_n = a_0$, then $J_p = \mathrm{id}$.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.pathEndpoint;Komorebi.PaperIII.Transport.exactPathTransport;Komorebi.PaperIII.Transport.exactPathTransport_eq_endpoint;Komorebi.PaperIII.Transport.exactPathTransport_loop`

Proof. Induct on the list of successive vertices and apply the cocycle in theorem 4.1 at each step. If the endpoint is the start, the endpoint transport is the identity. □

4.2 Exact packet coordinates

The graph model above resolves a fixed scalar matrix. Coordinate transport is a separate construction. Let X be an R-module, and let

$$\mathcal{P}^a = (P_i^a)_{i\in I_a}$$

be a finite complete family of pairwise orthogonal linear idempotents:

$$P_i^a P_j^a = \delta_{ij} P_i^a, \qquad \sum_{i \in I_a} P_i^a = I_X.$$

Define the exact packet coordinate module

$$\mathrm{Coord}(\mathcal{P}^a) = \{(x_i)_{i \in I_a} : P_i^a x_i = x_i\}.$$

Its analysis and synthesis maps are

$$\mathcal{A}_a(x) = (P_i^a x)_i, \qquad \mathcal{S}_a((x_i)_i) = \sum_i x_i.$$

Completeness and orthogonality give

$$\mathcal{S}_a \mathcal{A}_a = \mathrm{id}_X, \qquad \mathcal{A}_a \mathcal{S}_a = \mathrm{id}_{\mathrm{Coord}(\mathcal{P}^a)}.$$

Definition 4.3 (Canonical exact coordinate transport)**.** For two exact packet systems on the same ambient module, define

$$J^X_{b \leftarrow a} = \mathcal{A}_b \mathcal{S}_a : \mathrm{Coord}(\mathcal{P}^a) \longrightarrow \mathrm{Coord}(\mathcal{P}^b).$$

The arrow points from the source coordinate system a to the target coordinate system b.

Inherited formal contract: Papers I–II.
`Komorebi.PaperII.NestedRefinement.refineCoordinates;Komorebi.PaperII.NestedRefinement.coordinateRefinementEquiv`

Theorem 4.4 (Identity, inverse, and cocycle for canonical transport)**.** *For any exact packet systems a, b, c on X,*

$$J^X_{a \leftarrow a} = \mathrm{id}, \qquad \left(J^X_{b \leftarrow a}\right)^{-1} = J^X_{a \leftarrow b}, \qquad J^X_{c \leftarrow b} J^X_{b \leftarrow a} = J^X_{c \leftarrow a}.$$

Inherited formal contract: Papers I–II.
`Komorebi.PaperII.NestedRefinement.refineCoordinates_self;Komorebi.PaperII.NestedRefinement.coarsenCoordinates_refineCoordinates;Komorebi.PaperII.NestedRefinement.refineCoordinates_coarsenCoordinates;Komorebi.PaperII.NestedRefinement.refineCoordinates_comp;Komorebi.PaperII.NestedRefinement.coordinateRefinementEquiv_trans`

Proof. Insert $\mathcal{S}_b \mathcal{A}_b = \mathrm{id}_X$:

$$J^X_{c \leftarrow b} J^X_{b \leftarrow a} = \mathcal{A}_c \mathcal{S}_b \mathcal{A}_b \mathcal{S}_a = \mathcal{A}_c \mathcal{S}_a = J^X_{c \leftarrow a}.$$

The identity statement is $\mathcal{A}_a \mathcal{S}_a = \mathrm{id}$. Setting $c = a$ in the cocycle gives $J^X_{a \leftarrow b} J^X_{b \leftarrow a} = \mathrm{id}$; interchanging a and b gives the other composition, so the two transports are inverses. □

If $F : X \to Y$ is linear and $\mathcal{P}^a, \mathcal{P}^b$ are source packet systems while $\mathcal{Q}^r, \mathcal{Q}^s$ are target packet systems, put

$$\widehat{F}_{r \leftarrow a} = \mathcal{A}_r^Y F \mathcal{S}_a^X.$$

Theorem 4.5 (Canonical coordinate naturality)**.** *The exact coordinate representation preserves typed composition and satisfies*

$$J^Y_{s\leftarrow r}\widehat{F}_{r\leftarrow a} = \widehat{F}_{s\leftarrow b}J^X_{b\leftarrow a}.$$

Inherited formal contract: Papers I–II.
`Komorebi.PaperII.NestedRefinement.coordinateOperator_comp;Komorebi.PaperII.NestedRefinement.coordinateOperator_naturality`

Proof. Both sides reduce to $\mathcal{A}^Y_s F \mathcal{S}^X_a$ after inserting the appropriate synthesis–analysis identity. The composition law is the same calculation with the middle ambient module inserted. □

Corollary 4.6 (Canonical triangle holonomy is trivial)**.** *For every triple a, b, c,*

$$J^X_{a\leftarrow c}J^X_{c\leftarrow b}J^X_{b\leftarrow a} = \mathrm{id}.$$

Inherited formal contract: Papers I–II.
`Komorebi.PaperII.NestedRefinement.refineCoordinates_comp;Komorebi.PaperII.NestedRefinement.refineCoordinates_self`

Boundary 4.7 (Refinement is not reduction)*.* Although Paper II uses the name *refineCoordinates*, J is an equivalence between two complete coordinate presentations. It neither truncates packets nor projects to a smaller state space. Any lossy restriction, interpolation, or learned identification is extra data and cannot inherit theorem 4.4 without proof.

5 A distinct noncanonical connection API

5.1 The minimal compiled API

The formal Paper III API starts with deliberately weak assumptions. All chart transports act on one module M, so chart labels distinguish routes but not Lean types.

Definition 5.1 (Self-normalized noncanonical connection)**.** A minimal connection is a family

$$\Gamma_{b\leftarrow a} \in \mathrm{Aut}_R(M) \qquad (a, b \in B)$$

with only

$$\Gamma_{a\leftarrow a} = \mathrm{id}_M.$$

Neither reverse-edge inversion nor the cocycle law is a field of the structure.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.Connection`

At a triple (a, b, c), the compiled cocycle predicate is

$$\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} = \Gamma_{c\leftarrow a}.$$

Flatness means this equality holds for every triple. The formal multiplicative relative defect and triangle holonomy are

$$D^{\times}_{c,b,a} = (\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a})\Gamma^{-1}_{c\leftarrow a}, \qquad H_{c,b,a} = \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a}.$$

The defect is based at the target c; the holonomy is based at a.

Formal contract: Paper III v1.0.1.

`Komorebi.PaperIII.Transport.Connection.CocycleAt;Komorebi.PaperIII.Transport.Connection.Flat;Komorebi.PaperIII.Transport.Connection.cocycleDefect;Komorebi.PaperIII.Transport.Connection.triangleHolonomy`

Theorem 5.2 (Checked flatness consequences)**.** *For the minimal connection API, flatness implies*

$$D^{\times}_{c,b,a} = \mathrm{id}_M, \qquad H_{c,b,a} = \mathrm{id}_M$$

for every a, b, c*.*

Formal contract: Paper III v1.0.1.

`Komorebi.PaperIII.Transport.Connection.cocycleDefect_eq_refl_of_flat;Komorebi.PaperIII.Transport.Connection.triangleHolonomy_eq_refl_of_flat`

Proof. Flatness replaces the two-edge route in $D^{\times}$ by the direct route, which cancels its inverse. For H, flatness first collapses $a \to b \to c$ to $a \to c$, then collapses $a \to c \to a$ to the normalized transport $a \to a$. □

Proposition 5.3 (The minimal API admits nontrivial holonomy)**.** *There is a connection on three points over* $\mathbb{Q}$ *for which transport* $0 \to 1$ *is multiplication by* 2*, every other unequal-endpoint transport is the identity, and*

$$H_{2,1,0}(1) = 2.$$

In particular $H_{2,1,0} \neq \mathrm{id}$*.*

Formal contract: Paper III v1.0.1.

`Komorebi.PaperIII.Transport.twistedThreePointConnection;Komorebi.PaperIII.Transport.twisted_triangleHolonomy_apply_one;Komorebi.PaperIII.Transport.twisted_triangleHolonomy_ne_refl`

Boundary 5.4 (Why the minimal defect and loop are distinct). Without reverse-edge inversion, $\Gamma_{a \leftarrow c}$ need not equal $\Gamma^{-1}_{c \leftarrow a}$. Therefore the multiplicative relative defect $D^{\times}_{c,b,a}$ and triangle holonomy $H_{c,b,a}$ are different comparisons. The formal module proves that global flatness trivializes both; it does not identify them for an arbitrary connection.

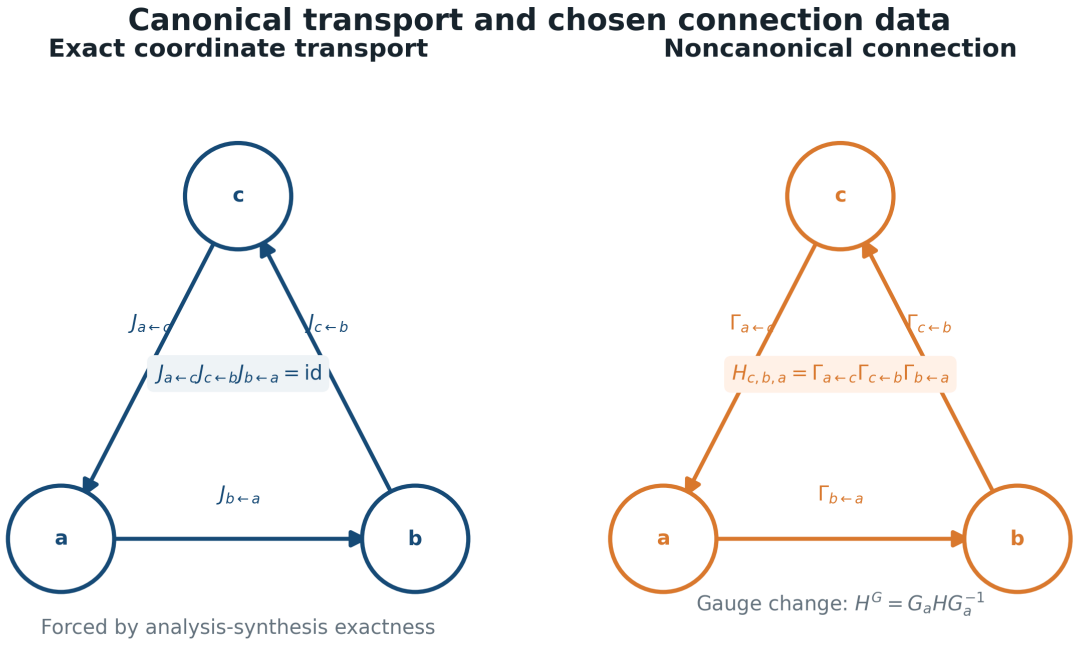


Figure 2. Exact coordinate transport and connection transport are distinct. The canonical loop telescopes to the identity. A chosen connection may carry nontrivial holonomy, which transforms by conjugation under gauge.

Figure 2 records this separation visually: the exact coordinate route telescopes, whereas the chosen connection route may carry a nonidentity loop.

5.2 A stronger reversible pair API

Let R be a commutative ring, let B be a type of charts or refinement states, and let each C_a be an R-module. The chart type is not assumed finite. The following definition is deliberately not derived from analysis and synthesis.

Definition 5.5 (Normalized reversible connection)**.** A normalized reversible connection is a family of linear equivalences

$$\Gamma_{b\leftarrow a} : C_a \xrightarrow{\sim} C_b \qquad (a, b \in B)$$

such that

$$\Gamma_{a\leftarrow a} = \mathrm{id}, \qquad \Gamma_{a\leftarrow b} = \Gamma_{b\leftarrow a}^{-1}.$$

No cocycle equation is included in the structure.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection`

This strengthens the homogeneous minimal connection with heterogeneous fibers and reverse-edge inversion. It remains additional data, not an alias for canonical coordinate transport.

Definition 5.6 (Calibration and flatness)**.** When the C_a are exact packet coordinates of one ambient module, the connection is *canonically calibrated* if

$$\Gamma_{b\leftarrow a} = J_{b\leftarrow a} \quad \text{for every } a, b.$$

It is *flat* if

$$\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} = \Gamma_{c\leftarrow a} \quad \text{for every } a, b, c.$$

The canonical connection is flat; a general flat connection is proved in theorem 6.3 to be gauge-equivalent to it.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.CocycleAt;Komorebi.PaperIII.Transport.ReversibleConnection.Flat;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalConnection;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalConnection_flat`

The words *calibrated* and *flat* have only the algebraic meanings displayed above.

Remark 5.7 (Graph-local variant). One may specify Γ_e only on the oriented edges of a chart graph, together with inverse maps on reversed edges. Path transport and loop holonomy below still make sense. A direct cocycle defect for (a, b, c), however, requires a declared direct transport from a to c. The pairwise API in definition 5.5 keeps this comparison total.

5.3 Cocycle defect and triangle holonomy

The commutative-ring hypothesis makes subtraction of linear maps available from this point through section 7.

Definition 5.8 (Cocycle defect)**.** For $a, b, c \in B$, define the additive cocycle defect

$$\Delta_{c,b,a} = \Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} - \Gamma_{c\leftarrow a} \in \mathrm{Hom}_R(C_a, C_c).$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.cocycleDefect`

The manuscript subscript order is target, intermediate, source. The Lean arguments are supplied in traversal order `(a,b,c)`.

Definition 5.9 (Triangle holonomy)**.** The triangle holonomy based at a is

$$H_{c,b,a} = \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} \in \mathrm{Aut}_R(C_a).$$

It traverses $a \to b \to c \to a$.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.triangleHolonomy;Komorebi.PaperIII.Transport.ReversibleConnection.triangleObstruction`

Holonomy is attached to the chosen connection Γ, never to an unspecified coordinate change.

Theorem 5.10 (Defect–holonomy identity)**.** *For every triple* (a, b, c)*,*

$$\boxed{H_{c,b,a} - \mathrm{id}_{C_a} = \Gamma_{a\leftarrow c}\Delta_{c,b,a}.}$$

Consequently,

$$\Delta_{c,b,a} = 0 \quad \Longleftrightarrow \quad H_{c,b,a} = \mathrm{id}_{C_a}.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.triangleHolonomy_sub_id;Komorebi.PaperIII.Transport.ReversibleConnection.cocycleDefect_eq_zero_iff_triangleHolonomy_eq_refl`

Proof. Using reversibility,

$$\begin{aligned}\Gamma_{a\leftarrow c}\Delta_{c,b,a} &= \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow b}\Gamma_{b\leftarrow a} - \Gamma_{a\leftarrow c}\Gamma_{c\leftarrow a}\\ &= H_{c,b,a} - \mathrm{id}_{C_a}.\end{aligned}$$

Left composition by the equivalence $\Gamma_{a\leftarrow c}$ is injective, which gives the equivalence of the two vanishing statements. □

Corollary 5.11 (Canonical transport has zero connection defect)**.** *If* $\Gamma_{b\leftarrow a} = J_{b\leftarrow a}$*, then*

$$\Delta_{c,b,a} = 0, \qquad H_{c,b,a} = \mathrm{id}$$

for all triples.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.canonicalTransport_zero_connectionDefect;Komorebi.PaperIII.Transport.ReversibleConnection.canonicalTransport_triangleHolonomy_eq_refl`

Proof. The defect vanishes by the canonical cocycle in theorem 4.4; apply theorem 5.10. □

5.4 Paths and endpoint independence

Definition 5.12 (Path-ordered connection transport)**.** For a vertex sequence $p = (a_0, a_1, \dots, a_n)$, define

$$\Gamma_p = \Gamma_{a_n\leftarrow a_{n-1}} \cdots \Gamma_{a_1\leftarrow a_0}, \qquad \Gamma_{(a_0)} = \mathrm{id}_{C_{a_0}}.$$

If $a_n = a_0$, then Γ_p is the holonomy of the loop p.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.pathTransport;Komorebi.PaperIII.Transport.ReversibleConnection.VertexPath;Komorebi.PaperIII.Transport.ReversibleConnection.indexedPathTransport`

Products are path-ordered: the first traversed edge acts on the right.

Proposition 5.13 (Flat pair transport collapses to endpoints)**.** *If* Γ *is flat, then every path satisfies*

$$\Gamma_p = \Gamma_{a_n\leftarrow a_0}.$$

In particular, every loop has identity holonomy.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.pathTransport_eq_direct_of_flat;Komorebi.PaperIII.Transport.ReversibleConnection.indexedPathTransport_eq_direct_of_flat;Komorebi.PaperIII.Transport.ReversibleConnection.indexedLoopTransport_eq_refl_of_flat`

Proof. Induct on the path length. Appending a_{n+1} left-multiplies the inductive expression by $\Gamma_{a_{n+1}\leftarrow a_n}$, and flatness replaces the two factors by $\Gamma_{a_{n+1}\leftarrow a_0}$. For a loop, normalization gives $\Gamma_{a_0\leftarrow a_0} = \mathrm{id}$. □

Remark 5.14. The converse *trivial holonomy implies flatness* depends on which loops are present in a graph-local API. In the complete pair API, triviality of every triangle holonomy is equivalent to flatness by theorem 5.10. No topological simple-connectivity theorem is being invoked.

6 Gauge covariance and comparison with canonical transport

Definition 6.1 (Gauge change)**.** A gauge is a family of automorphisms $G_a \in \mathrm{Aut}_R(C_a)$. It acts on a connection by

$$\Gamma^G_{b\leftarrow a} = G_b \Gamma_{b\leftarrow a} G_a^{-1}.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.Gauge;Komorebi.PaperIII.Transport.ReversibleConnection.gauge`

A gauge changes coordinate representatives at each chart; it does not change the chart graph.

Theorem 6.2 (Gauge covariance)**.** *The defect and triangle holonomy transform as*

$$\Delta^G_{c,b,a} = G_c \Delta_{c,b,a} G_a^{-1}, \qquad H^G_{c,b,a} = G_a H_{c,b,a} G_a^{-1}.$$

Thus flatness and triviality of triangle holonomy are gauge invariant.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.gauge_cocycleDefect;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_triangleHolonomy;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_flat_iff;Komorebi.PaperIII.Transport.ReversibleConnection.gauge_triangleHolonomy_eq_refl_iff`

Proof. Substitute the definition of Γ^G. Adjacent factors $G_b^{-1}G_b$ and $G_c^{-1}G_c$ cancel. In the defect, both the composite and direct terms acquire the same outer factors G_c and G_a^{-1}. In the loop, all inner factors cancel and only conjugation at the base remains. □

Theorem 6.3 (Flat connections are gauges of the canonical pair transport)**.** *Suppose the C_a are exact packet-coordinate modules for one ambient module, so the canonical $J_{b\leftarrow a}$ is available. If Γ is a flat normalized reversible pair connection, choose $o \in B$ and set*

$$G_a = \Gamma_{a\leftarrow o} J_{o\leftarrow a}.$$

Then every pair transport satisfies

$$\boxed{\Gamma_{b\leftarrow a} = G_b J_{b\leftarrow a} G_a^{-1}.}$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.ReversibleConnection.flatGauge;Komorebi.PaperIII.Transport.ReversibleConnection.flat_isGaugeOfCanonical`

Proof. Reversibility gives $G_a^{-1} = J_{a\leftarrow o}\Gamma_{o\leftarrow a}$. Therefore

$$\begin{aligned} G_b J_{b\leftarrow a} G_a^{-1} &= \Gamma_{b\leftarrow o} J_{o\leftarrow b} J_{b\leftarrow a} J_{a\leftarrow o} \Gamma_{o\leftarrow a} \\ &= \Gamma_{b\leftarrow o}\Gamma_{o\leftarrow a} \\ &= \Gamma_{b\leftarrow a}. \end{aligned}$$

The first reduction uses the canonical cocycle twice; the second uses flatness of Γ. □

Boundary 6.4 (Flat does not mean calibrated). The theorem produces a gauge relating Γ to J. It does not prove $\Gamma = J$ before that gauge is chosen. Conversely, nontrivial triangle holonomy cannot be removed by a gauge, because conjugation cannot turn a nonidentity automorphism into the identity.

7 Obstruction packets: local certificates and holonomy blocks

7.1 The generic compiled certificate

Let Candidate, Packet, and Residual be arbitrary types, with a distinguished $0 \in$ Residual, and let

$$r : \mathsf{Candidate} \longrightarrow \mathsf{Packet} \longrightarrow \mathsf{Residual}$$

be a residual function.

Definition 7.1 (Candidate-specific solution and obstruction packet)**.** A candidate x is a solution when

$$\mathrm{IsSolution}_r(x) \quad \Longleftrightarrow \quad \forall p,\ r(x,p) = 0.$$

An obstruction packet for x is a pair consisting of a named packet p and a certificate $r(x,p) \neq 0$.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.NonlinearObstructions.IsSolution;Komorebi.PaperIII.NonlinearObstructions.ObstructionPacket`

Theorem 7.2 (Exact logical force of a local obstruction)**.** *Every obstruction packet for x proves that x is not a solution, and*

$$\neg\,\mathrm{IsSolution}_r(x) \quad \Longleftrightarrow \quad \textit{there exists an obstruction packet for } x.$$

The reverse implication from failure to a packet witness is classical and does not require the packet type to be finite.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.NonlinearObstructions.ObstructionPacket.not_isSolution;Komorebi.PaperIII.NonlinearObstructions.obstructionPacket_of_not_isSolution;Komorebi.PaperIII.NonlinearObstructions.not_isSolution_iff_nonempty_obstructionPacket`

Proof. A purported solution would make the residual vanish at the certified packet, contradicting its nonzero field. Conversely, if no packet carries a nonzero residual, then every residual is zero and the candidate satisfies the definition of solution. □

Boundary 7.3 (Candidate-specific means candidate-specific). The theorem rules out the named x. It neither constructs a solution nor shows that every candidate is obstructed. It provides no minimality, stability, computability, or classification theorem for the packet witness.

7.2 Holonomy-block specialization

For a reversible connection, the generic residual may be specialized at a fixed triangle by taking a packet label to be a pair (μ, ν) and the residual to be the corresponding block of $H - \mathrm{id}$. A block is therefore a packet residual; when it is nonzero, the label together with a proof of nonvanishing is an obstruction-packet certificate in the sense of definition 7.1. The reconstruction theorem below is the extra statement needed to turn vanishing of all those local residuals into the cocycle equation at that triangle.

Fix a triangle (a, b, c) and resolve the base fiber C_a by a finite complete family of idempotents $(\Pi^a_\mu)_{\mu \in I_a}$. Orthogonality may also hold, as it does for the exact range-coordinate projectors below, but the compiled reconstruction theorem does not require it. In exact range coordinates, Π^a_μ is simply the projector that retains component μ and zeros the other components.

Definition 7.4 (Holonomy obstruction and packet residuals)**.** Define

$$\mathrm{Ob}_{c,b,a} = H_{c,b,a} - \mathrm{id}_{C_a}, \qquad \mathrm{Ob}^{\mu\nu}_{c,b,a} = \Pi^a_\mu \, \mathrm{Ob}_{c,b,a} \, \Pi^a_\nu.$$

The map $\mathrm{Ob}^{\mu\nu}_{c,b,a}$ is the obstruction block, or packet residual, from input packet ν to output packet μ. If it is nonzero, then (μ, ν) and that nonvanishing proof form an obstruction-packet certificate. Diagonal blocks detect within-packet holonomy; off-diagonal blocks detect holonomy-induced packet mixing.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.HolonomyBlocks.PacketResolution;Komorebi.PaperIII.HolonomyBlocks.PacketResolution.block`

An obstruction-packet certificate here is a nonzero block of a precisely specified holonomy defect together with its packet-pair label; it is not merely a heuristic label for a large residual.

Theorem 7.5 (Exact obstruction-block reconstruction and detection)**.** *The total obstruction has the exact finite resolution*

$$\boxed{\mathrm{Ob}_{c,b,a} = \sum_{\mu,\nu \in I_a} \mathrm{Ob}^{\mu\nu}_{c,b,a}.}$$

Moreover, the following are equivalent:

(i) $\Delta_{c,b,a} = 0$*;*

(ii) $H_{c,b,a} = \mathrm{id}$*;*

(iii) $\mathrm{Ob}_{c,b,a} = 0$*;*

(iv) $\mathrm{Ob}^{\mu\nu}_{c,b,a} = 0$ *for every* μ, ν*.*

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.HolonomyBlocks.obstruction_sum_blocks;Komorebi.PaperIII.HolonomyBlocks.obstructionBlocks_zero_iff;Komorebi.PaperIII.HolonomyBlocks.obstructionDetection_equivalences`

Proof. Insert completeness on both sides:

$$\mathrm{Ob} = \left(\sum_{\mu} \Pi^a_\mu\right) \mathrm{Ob} \left(\sum_{\nu} \Pi^a_\nu\right) = \sum_{\mu,\nu} \Pi^a_\mu \,\mathrm{Ob}\, \Pi^a_\nu.$$

Thus (iv) implies (iii), and (iii) implies (iv) by composition with the projectors. Conditions (ii) and (iii) are equivalent by the definition of Ob and additive cancellation. Conditions (i) and (ii) are equivalent by theorem 5.10. □

Figure 3 depicts the two-sided finite block resolution behind the reconstruction identity and its off-diagonal witnesses.

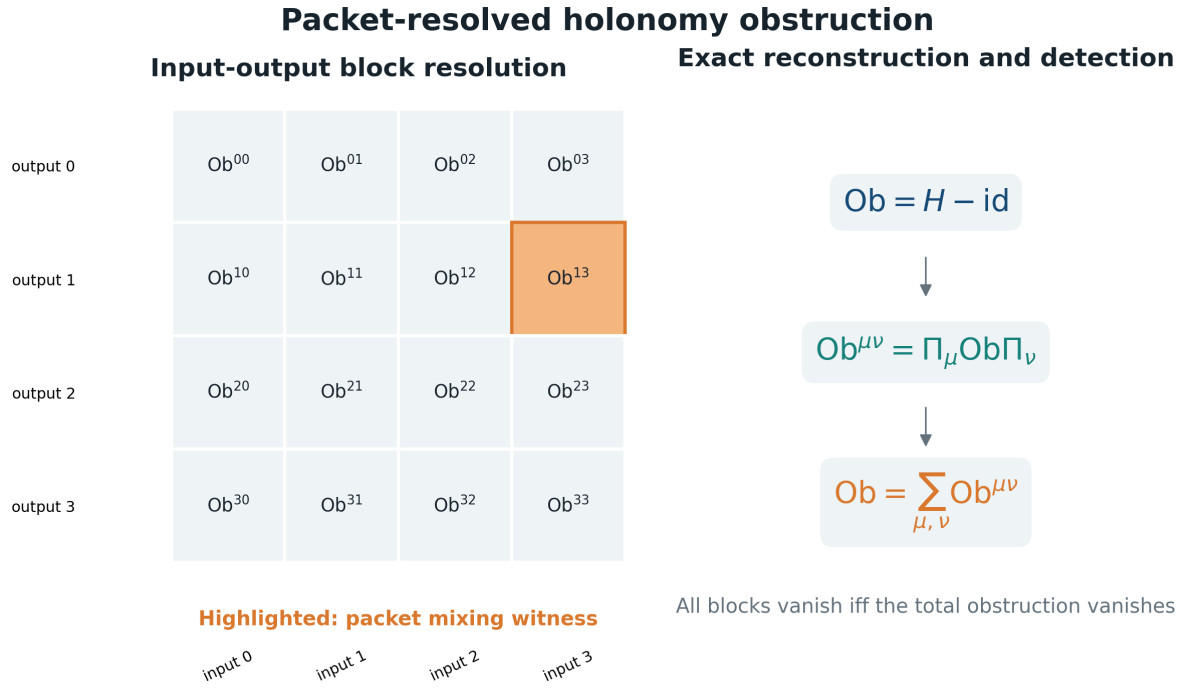


Figure 3. A finite complete packet resolution reconstructs the total holonomy obstruction from its input-output blocks. A nonzero off-diagonal block is an exact packet-mixing witness.

Corollary 7.6 (Packet-supported witness)**.** *If* $\mathrm{Ob}^{\mu\nu}_{c,b,a} \neq 0$*, then there is a vector* x *supported in packet* ν *such that*

$$\Pi^a_\mu \,\mathrm{Ob}_{c,b,a}\, x \neq 0.$$

Such an x *is an exact finite witness of nonflatness.*

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.HolonomyBlocks.exists_packetWitness_of_obstructionBlock_ne_zero`

Proof. Classically choose y on which the nonzero linear map $\Pi^a_\mu \,\mathrm{Ob}\, \Pi^a_\nu$ is nonzero and set $x = \Pi^a_\nu y$. Idempotence gives $\Pi^a_\nu x = x$, and the required output is the chosen nonzero value. □

7.3 Gauge behavior of packet blocks

Vanishing or nonvanishing of the total triangle obstruction is gauge invariant by theorem 6.2, but a fixed packet pattern need not be. A general G_a can mix the projectors Π^a_μ.

Proposition 7.7 (Packet-preserving gauge covariance). *If $G_a\Pi^a_\mu = \Pi^a_\mu G_a$ for every μ, then*

$$\left(\mathrm{Ob}^G\right)^{\mu\nu} = G_a\,\mathrm{Ob}^{\mu\nu}\,G_a^{-1}.$$

Consequently the zero/nonzero status of each obstruction block is preserved by packet-preserving gauges.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.HolonomyBlocks.obstructionBlock_gauge_of_commutes;Komorebi.PaperIII.HolonomyBlocks.obstructionBlock_gauge_eq_zero_iff`

Proof. Use $\mathrm{Ob}^G = G_a\,\mathrm{Ob}\,G_a^{-1}$ and commute G_a and G_a^{-1} through the two packet projectors. Composition by an equivalence is injective in both directions. □

Boundary 7.8 (Packet patterns are not absolute). Without the commutation hypothesis in proposition 7.7, only the vanishing of the total obstruction is gauge invariant. The location of individual nonzero blocks may change under a gauge that mixes packets.

8 Packet-supported products and holonomy walks

The word-to-walk mechanism is not special to powers of one matrix. Let V be one finite packet type, let R be a semiring, and let $B_1, \dots, B_m \in R^{V\times V}$ be a finite sequence of possibly different packet matrices. Rows are targets and columns are sources as before.

Proposition 8.1 (Heterogeneous packet-path expansion). *For $i, j \in V$ and $m \geq 1$, set $i_0 = i$ and $i_m = j$. Then*

$$\boxed{(B_m \cdots B_1)_{ji} = \sum_{(i_1,\dots,i_{m-1})\in V^{m-1}} B_m[i_m, i_{m-1}]\cdots B_1[i_1, i_0].}$$

When $m = 1$, the sum is over the unique empty tuple and its summand is $B_1[j, i]$. For $m = 0$, the product is the identity and the entry is the Kronecker delta δ_{ji}. If every $B_r[q, p]$ vanishes outside a declared relation $\mathcal{R}_r(p, q)$, then the endpoint entry is zero whenever there is no packet path compatible with the successive relations.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.EdgeOperators.heterogeneousProduct;Komorebi.PaperIII.EdgeOperators.heterogeneousPathSum;Komorebi.PaperIII.EdgeOperators.heterogeneousPacketPathExpansion;Komorebi.PaperIII.EdgeOperators.SupportedStep;Komorebi.PaperIII.EdgeOperators.SupportedPath;Komorebi.PaperIII.EdgeOperators.heterogeneousProduct_apply_eq_zero_of_no_supportedPath`

Proof. Expand each matrix product entry and distribute the finite sums. Induction on the factor list yields the displayed path sum, with the empty list evaluated as the identity matrix. If no supported intermediate packet exists, every summand contains an entry that is zero by its declared support relation. □

Boundary 8.2 (Changing packet types along a chain). The compiled theorems `heterogeneousPacketPathExpansion_two` and `heterogeneousPacketPathExpansion_three` allow different packet types at the successive stages of two- and three-factor products. The arbitrary-length theorem above permits different factors but uses one common finite packet type V. A fully dependent arbitrary-length chain of changing modules and packet types remains outside the version 1.0.1 contract.

Remark 8.3 (Checked specialization). When every B_r is one repeated endomorphism T, the recursive packet-power specialization is inherited from Paper I.

Inherited formal contract: Papers I–II.
`Komorebi.PacketNaturalityCompletion.Packets.transferBlock_pow_eq_packetPathSum`

For a loop represented on one common finite scalar packet-coordinate type, the proposition resolves its successive transport matrices into packet walks. If R is a ring, as in the reversible-connection API, then $(H - I)_{ji} = H_{ji} - \delta_{ji}$; in particular, off-diagonal entries are unchanged. Thus a nonzero off-diagonal holonomy obstruction in this common-index model requires a transport word along the closed chart loop whose packet path runs from its input packet to its output packet. This conclusion is conditional on declared block supports for the connection matrices; an arbitrary connection need not respect the transfer graph from section 2.

9 A minimal exact example

Example 9.1 (A scalar triangle)**.** Let $B = \{0, 1, 2\}$ and $C_0 = C_1 = C_2 = \Bbbk$ for a field $\Bbbk$. Choose $u, v, w \in \Bbbk^\times$ and set

$$\Gamma_{1\leftarrow 0}(x) = ux, \qquad \Gamma_{2\leftarrow 1}(x) = vx, \qquad \Gamma_{2\leftarrow 0}(x) = wx,$$

with reverse transports given by inverse scalars. Then

$$\Delta_{2,1,0}(x) = (vu - w)x,$$

and the triangle holonomy based at 0 is

$$H_{2,1,0}(x) = w^{-1}vux.$$

Therefore

$$H_{2,1,0} - \mathrm{id} = w^{-1}\Delta_{2,1,0}.$$

The connection is flat on this triangle exactly when $vu = w$. If the three coordinate systems are literally the same and canonical transport is used, then $u = v = w = 1$ and the obstruction vanishes. Choosing arbitrary nonzero u, v, w is a connection choice, not a consequence of changing exact packet coordinates.

Example 9.2 (A two-step transfer walk)**.** Let a packet graph have edges $e : 0 \to 1$ and $f : 1 \to 2$, with no other edges. Then

$$T = L_e + L_f, \qquad (T^2)_{20} = c(f)c(e).$$

The products L_eL_e, L_eL_f, and L_fL_f vanish by typed mismatch; only L_fL_e, corresponding to the walk $0 \to 1 \to 2$, contributes to the $(2,0)$ entry. This example also fixes the noncommutative coefficient order.

10 Canonical coordinates for arbitrary maps

The analysis and synthesis identities are identities of functions, not only of linear maps. Consequently exact coordinate change does not become curved merely because the represented operation is nonlinear.

Proposition 10.1 (Set-level exact naturality)**.** *Let $\mathcal{A}_a^X : X \xrightarrow{\sim} X_a$ and $\mathcal{A}_b^X : X \xrightarrow{\sim} X_b$ be exact coordinate equivalences for X, with inverses $\mathcal{S}_a^X$ and $\mathcal{S}_b^X$. Likewise, let $\mathcal{A}_r^Y : Y \xrightarrow{\sim} Y_r$ and $\mathcal{A}_s^Y : Y \xrightarrow{\sim} Y_s$ be exact coordinate equivalences for Y, with inverses $\mathcal{S}_r^Y$ and $\mathcal{S}_s^Y$. Let $F : X \to Y$ be any function, with no linearity or differentiability hypothesis. Set*

$$J_{b\leftarrow a}^X = \mathcal{A}_b^X \circ \mathcal{S}_a^X, \qquad J_{s\leftarrow r}^Y = \mathcal{A}_s^Y \circ \mathcal{S}_r^Y,$$

and define

$$\widehat{F}_{r\leftarrow a} = \mathcal{A}_r^Y \circ F \circ \mathcal{S}_a^X.$$

Then

$$J_{s\leftarrow r}^Y \circ \widehat{F}_{r\leftarrow a} = \widehat{F}_{s\leftarrow b} \circ J_{b\leftarrow a}^X.$$

If $\mathcal{A}_t^Z : Z \xrightarrow{\sim} Z_t$ is another exact coordinate equivalence and $G : Y \to Z$ is any function, then exact coordinate representation also preserves composition:

$$\widehat{G \circ F}_{t\leftarrow a} = \widehat{G}_{t\leftarrow r} \circ \widehat{F}_{r\leftarrow a}.$$

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.Transport.exactCoordinateMap;Komorebi.PaperIII.Transport.exactCoordinateMap_comp;Komorebi.PaperIII.Transport.exactCoordinateMap_naturality`

Proof. Both sides of the naturality equation reduce to $\mathcal{A}_s^Y \circ F \circ \mathcal{S}_a^X$ using $\mathcal{S} \circ \mathcal{A} = \mathrm{id}$ in X and Y. For composition, the middle $\mathcal{S}^Y \circ \mathcal{A}^Y$ cancels in the same way. □

Boundary 10.2 (Where nonlinear defects can enter). A nonlinear *coordinate* defect cannot be blamed on the exact canonical J of definition 4.3; proposition 10.1 makes that defect identically zero. Nonzero defects require extra structure such as a noncanonical connection, truncation, approximation, restricted domains, or failure of an operation to respect a declared packet model.

11 Nonlinear operation trees: locality and the formalization frontier

11.1 The compiled unary–binary foundation

Definition 11.1 (Finite operation-tree grammar)**.** For types of input labels I, constants K, unary symbols U, and binary symbols B, the released formal grammar has four constructors:

$$\begin{aligned} t ::= \ & \mathrm{input}(i) \mid \mathrm{constant}(k) \mid \mathrm{unary}(u, t) \\ & \mid \mathrm{binary}(b, t_1, t_2). \end{aligned}$$

Given an arbitrary value type Y, evaluation takes completely explicit maps

$$I \to Y, \qquad K \to Y, \qquad U \to Y \to Y, \qquad B \to Y \to Y \to Y.$$

No algebraic, continuity, or differentiability law is imposed on these semantics.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.NonlinearObstructions.OperationTree;Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval`

When I has decidable equality, let $\mathrm{inputs}(t)$ be the finite set of input labels actually occurring in t.

Theorem 11.2 (Operation-tree input locality)**.** *Let $\eta_1, \eta_2 : I \to Y$ be two input environments. If*

$$\eta_1(i) = \eta_2(i) \qquad \text{for every } i \in \mathrm{inputs}(t),$$

then the two evaluations of t are equal under the same constant, unary, and binary semantics. In particular, if $\mathrm{inputs}(t) = \varnothing$, evaluation is independent of the input environment.

Formal contract: Paper III v1.0.1.
`Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval_congr_on_inputs;Komorebi.PaperIII.NonlinearObstructions.OperationTree.eval_eq_of_inputs_eq_empty`

Proof. Structural induction on t. The input case is the agreement hypothesis; the constant case is reflexive; the unary case applies the induction hypothesis to its child; and the binary case applies it independently to the left and right subtrees, whose input sets form a union. □

Boundary 11.3 (Exact force of locality). The theorem says evaluation reads only syntactically named inputs. It does not prove that the operations are linear or nonlinear in any analytic sense, that the tree is stable, that two different trees are equivalent, or that a solution to a tree equation exists.

11.2 Typed colored extension beyond the finite contract

Definition 11.4 (Finitary colored operation signature)**.** A finitary colored signature consists of:

- a finite type V of colors;
- a type Σ of operation symbols;
- an arity $r(\sigma) \in \mathbb{N}$ for every $\sigma \in \Sigma$;

- input colors $\mathrm{in}_\sigma : \mathrm{Fin}(r(\sigma)) \to V$;
- an output color $\mathrm{out}(\sigma) \in V$.

A realization assigns a space X_v to each color and a map

$$\Phi_\sigma : \prod_{q \in \mathrm{Fin}(r(\sigma))} X_{\mathrm{in}_\sigma(q)} \longrightarrow X_{\mathrm{out}(\sigma)}.$$

Outside the finite formal contract. The intended Lean target is a dependent inductive typed tree; no such declaration is part of the version 1.0.1 completion contract.

Definition 11.5 (Typed operation tree and evaluation)**.** A typed operation tree is recursively either a leaf of declared color or a node labeled by σ, with one correctly colored child for each input slot of σ. Evaluation is recursive:

$$\mathrm{ev}(\mathrm{node}(\sigma; t_1, \ldots, t_{r(\sigma)})) = \Phi_\sigma(\mathrm{ev}(t_1), \ldots, \mathrm{ev}(t_{r(\sigma)})).$$

Outside the finite formal contract. Candidate API: `Komorebi.PaperIII.NonlinearObstructions.ColoredOperationTree` and `ColoredOperationTree.eval`. This is stronger than the compiled uncolored unary–binary grammar.

11.3 Packet labelings and the multilinear first target

Suppose every X_v has a finite complete packet family $(P_i^v)_{i \in I_v}$. A packet labeling assigns an admissible packet index to each edge or node of a typed tree. For multilinear Φ_σ, inserting

$$\mathrm{id}_{X_v} = \sum_{i \in I_v} P_i^v$$

at every input and output yields a finite sum over packet-labeled trees. This is the branching analogue of inserting intermediate projectors in proposition 8.1.

Formalization target 11.6 (Finite multilinear tree expansion)*.* The first milestone for the nonlinear module is an exact theorem saying that the evaluation of a finite tree of multilinear nodes equals the finite sum of its packet-resolved labeled evaluations. Required proof obligations include:

(a) a dependent well-typed tree syntax;

(b) finite packet-label types at every color;

(c) an explicit convention for nullary and unary nodes;

(d) multilinearity in every input slot;

(e) preservation of input-slot order without assuming commutativity of operations.

Outside the finite formal contract. Candidate target: `Komorebi.PaperIII.NonlinearObstructions.ColoredOperationTree.eval_eq_sum_packetLabellings`.

11.4 Connection defects at nonlinear nodes

For coordinate models of the source and target colors, an additive target coordinate space, and a chosen noncanonical connection, a pointwise node defect can be defined by

$$\mathfrak{D}^{\Gamma}_{b,a}(\Phi) = \Gamma^{Y}_{b\leftarrow a} \circ \widehat{\Phi}_a - \widehat{\Phi}_b \circ \left(\prod_q \Gamma^{X_q}_{b\leftarrow a} \right).$$

This expression compares two *chosen* routes. With canonical exact transport in every slot it vanishes by proposition 10.1.

Formalization target 11.7 (Operation-tree defect propagation). For multilinear nodes, one expects an exact finite expansion obtained by replacing one or more nodes by their defects and transporting the remaining branches. For smooth nonlinear nodes, the corresponding chain rule is indexed by rooted partitions and Faà di Bruno trees. A formal theorem will require a fixed differentiability category, base points, derivative conventions, symmetry factors, and remainder policy.

Outside the finite formal contract. Tree-level defect identities, higher chain rules, and nonlinear norm bounds are outside the finite Lean contract. Ordinary mathematical results in these directions appear in Part II under their stated hypotheses.

Boundary 11.8 (Analytic and geometric exclusions). The finite formal contract does not include convergence of an infinite tree series, existence of derivatives, a Banach-space Faà di Bruno theorem, a Baker–Campbell–Hausdorff expansion, a continuum limit, or an identification of algebraic holonomy with curvature of a smooth bundle. Part II supplies ordinary proofs for its explicit nonlinear and analytic statements. This edition does not derive a connection from data, optimization, physics, or probability.

12 Formalization ledger for version 1.0.1

Topic	Status	Declaration or roadmap target	Exact boundary
Packet matrix and edge operator	Lean-backed III	`PacketGraph.packetMatrix`; `EdgeSystem.edgeOp`	Finite scalar matrices over a semiring; one scalar coordinate per packet.
Edge support and pair dichotomy	Lean-backed III	`edgeOp_eq_zero_off_support`; `edgeOp_mul_edgeOp_eq_zero`; `edgeOp_mul_edgeOp_of_composable`	Broken interfaces vanish; matched pairs collapse to the endpoint block with coefficient in operator order.
Typed word evaluation	Lean-backed III	`Composable`; `wordOp`; `wordOp_append`; `wordOp_eq_zero_of_not_composable`	List order is execution order; concatenation reverses operator-product order; nonwalks vanish.

Topic	Status	Declaration or roadmap target	Exact boundary
Finite word and walk expansions	Lean-backed III	`explicitWordExpansion`; `walkExpansion`; `wordExpansion_eq_explicitWordExpansion`; `transferOp_pow_eq_walkExpansion`	Actual finite tuple sum and zero-filtered composable-walk sum; finite graph and packet types.
Endpoint coefficient walk sum	Lean-backed III	`reverseCoefficient`; `Connects`; `transferOp_pow_apply_eq_sum_walks`	Exact scalar endpoint entry, including the empty-walk case and noncommutative coefficient order.
Fixed reachability and orbit support	Lean-backed III; inherited hull bridge	`transferOp_pow_apply_eq_zero_of_no_walk`; `transferOp_pow_mulVec_apply_eq_zero_of_unreachable_support`	Finite powers and supported seeds; the orbit-hull span conclusion uses the frozen Paper I contract.
Exact frame transport	Lean-backed III	`exactTransport_cocycle`; `exactPathTransport_eq_endpoint`; `exactPathTransport_loop`	One frame family on a single module; arbitrary finite paths telescope and closed paths are trivial.
Exact coordinate cocycle	Inherited Lean-backed II	`NestedRefinement.refineCoordinates_comp`; `NestedRefinement.coordinateRefinementEquiv_trans`	Complete finite projector coordinates, not reduction.
Typed coordinate naturality	Inherited Lean-backed II	`NestedRefinement.coordinateOperator_comp`; `NestedRefinement.coordinateOperator_naturality`	Linear maps over semiring modules.
Minimal connection API	Lean-backed III	`Transport.Connection`; `cocycleDefect`; `triangleHolonomy`; `triangleHolonomy_eq_refl_of_flat`	Only self-normalization is assumed; reverse inversion and the cocycle are not structure fields.
Twisted three-point loop	Lean-backed III	`twistedThreePointConnection`; `twisted_triangleHolonomy_ne_refl`	Concrete rational connection; the selected loop sends 1 to 2.
Reversible additive defect	Lean-backed III	`ReversibleConnection`; `triangleHolonomy_sub_id`; `cocycleDefect_eq_zero_iff_triangleHolonomy_eq_refl`	Heterogeneous module fibers, reverse-edge inversion, and an additive defect over a commutative ring.

Topic	Status	Declaration or roadmap target	Exact boundary
Reversible finite paths	Lean-backed III	`pathTransport_eq_direct_of_flat`; `indexedPathTransport_eq_direct_of_flat`; `indexedLoopTransport_eq_refl_of_flat`	Finite pair-connection paths collapse to endpoints under algebraic flatness.
Gauge covariance	Lean-backed III	`gauge_cocycleDefect`; `gauge_triangleHolonomy`; `gauge_flat_iff`	Fiberwise algebraic conjugation; no smooth gauge theory.
Flat versus canonical transport	Lean-backed III	`flatGauge`; `flat_isGaugeOfCanonical`	A chosen base chart and ambient frames exhibit every flat pair connection as a gauge transform of canonical transport.
Operation-tree locality	Lean-backed III	`OperationTree`; `OperationTree.eval`; `OperationTree.eval_congr_on_inputs`	Arbitrary unary and binary semantics; only dependence on named inputs is proved.
Generic obstruction packet	Lean-backed III	`ObstructionPacket.not_isSolution`; `not_isSolution_iff_nonempty_obstructionPacket`	Exact candidate-specific logical certificate; no global classification.
Holonomy block obstruction	Lean-backed III	`obstruction_sum_blocks`; `obstructionBlocks_zero_iff`; `obstructionDetection_equivalences`	Finite complete idempotent packet family on the base fiber; orthogonality is not required by the compiled reconstruction.
Packet witness and block gauge	Lean-backed III	`exists_packetWitness_of_obstructionBlock_ne_zero`; `obstructionBlock_gauge_of_commutes`; `obstructionBlock_gauge_eq_zero_iff`	Witness extraction is classical; blockwise covariance requires the gauge to commute with every packet projector.
Heterogeneous matrix paths	Lean-backed III	`heterogeneousPacketPathExpansion`; `heterogeneousProduct_apply_eq_zero_of_no_supportedPath`	Arbitrarily many different matrices over one common finite packet type; fully typed changing-index chains compile only at lengths two and three.

Topic	Status	Declaration or roadmap target	Exact boundary
Arbitrary-function coordinates	Lean-backed III	`exactCoordinateMap_comp`; `exactCoordinateMap_naturality`	Exact set-level equivalences and arbitrary functions; no linearity or differentiability assumption.
Colored and smooth trees	Roadmap only	roadmap `ColoredOperationTree.eval_eq_sum_packetLabellings`	Dependent colors, multilinear packet expansion, derivatives, and Faà di Bruno identities remain outside the contract.

12.1 Kernel-backed Paper III surface

The authoritative declaration-level contract is `formalization/lean/LEAN_THEOREM_MAP_v1.0.1.md`. It groups the compiled surface as follows.

- **EdgeOperators and EdgeExtensions:** scalar packet matrices, typed edge and word support, finite word/walk expansions, endpoint coefficient sums, fixed-length reachability, orbit support, and common-index heterogeneous products with declared supports.
- **Transport and ReversibleConnections:** exact frame transport, the minimal connection, heterogeneous reversible pair transport, additive defect and triangle holonomy, finite paths, canonical connections, and the fiberwise gauge action.
- **CoordinateNaturality and HolonomyBlocks:** composition and naturality for arbitrary functions between exact coordinate systems, finite two-sided block reconstruction, equivalent obstruction criteria, packet-supported witnesses, and packet-preserving block gauges.
- **NonlinearObstructions:** finite unary–binary operation trees, input locality, and candidate-specific obstruction certificates.

The release verification report records the checker-generated declaration and audit totals. The parity gate requires every fully qualified Paper III target printed beside a statement to occur in the theorem map, rejects stale or duplicate targets, and checks it against the compiled environment. `KernelAudit.lean` reports dependencies for the mapped and supporting theorems, and its fail-closed gate scans every compiled `Komorebi.PaperIII` declaration. The four inherited statement contracts remain identified separately by their Papers I–II names.

Boundary 12.1 (No scope promotion by prose). The displayed endpoint coefficient formula, reachability results, reversible connection theorems, additive defect–holonomy identity, gauge covariance, holonomy-block reconstruction, packet witnesses, common-index heterogeneous path expansion, and arbitrary-function coordinate naturality are all part of the version 1.0.1 kernel-backed surface. What remains outside is the dependent fiber-valued edge-system generalization, a fully dependent arbitrary-length chain whose module and packet type change at every stage, dependent colored

operation trees, packet-labeled multilinear tree expansions, nonlinear tree-defect and Faà di Bruno identities, and the analytic and geometric extensions listed above.

13 The finite contract and the nonlinear transition

Typed packet support turns the expansion of a transfer power into a walk calculus: incompatible words vanish for structural reasons, not because of numerical cancellation. Exact packet-coordinate transport is even more rigid. It is the canonical analysis–synthesis equivalence and therefore satisfies its cocycle law exactly; every finite canonical loop telescopes to the identity. Holonomy enters only after a separate connection is supplied. The kernel-backed minimal API proves that flatness kills both the relative defect and holonomy, and it exhibits a nontrivial rational loop. Under the kernel-backed reversible API, a nonzero additive triangle defect is equivalent to nonidentity holonomy, transforms covariantly under gauge, and is detected exactly by a finite family of packet blocks. Finite flat paths collapse to their endpoints, and every flat reversible pair connection is explicitly gauge-equivalent to canonical transport through a chosen base chart.

These facts define the boundary of the finite formal contract. At that boundary, exact coordinates preserve composition and satisfy naturality for arbitrary functions in Paper III, while the typed linear packet-coordinate counterpart remains inherited from Paper II. Nontrivial nonlinear defects must therefore be attached to additional modeling choices or to operation-specific packet compatibility. Colored operation trees, multilinear packet labelings, and Faà di Bruno expansions remain outside this finite Lean contract. Part II develops the corresponding ordinary mathematics, including packet jets, rooted tree expansions, and explicit nonlinear obstruction theorems.

Part II

Nonlinear packet calculus, obstruction complexes, and holonomy

The strengthened development below moves from exact arbitrary-function coordinates to differentiable interactions, specified correction problems, and finite or smooth transport. Its ordinary proofs retain their own hypotheses. They are not asserted to belong to the finite Lean contract of Part I.

The notational change from $\mathcal{A}, \mathcal{S}$ to $\mathcal{A}, \mathcal{S}$ is typographical: both denote analysis and synthesis for the declared packet system. A connection is still additional data. Smooth canonical transport is constructed from a moving projector family, whereas the algebraic exact coordinate transport in Part I compares fixed exact descriptions of one ambient object.

14 Introduction

A finite family of pairwise annihilating idempotents gives exact coordinates. If

$$P_\alpha P_\beta = 0 \quad \text{for } \alpha \neq \beta, \qquad \sum_\alpha P_\alpha = I,$$

then every vector is the sum of its packet components. Linear operators become block matrices in these coordinates. A nonlinear map cannot be reconstructed from its diagonal restrictions, since mixed interactions between packet components are usually present. It can still be represented exactly if its input is the complete packet tuple.

This article develops the nonlinear representation and the obstruction theory that follows from it. Exact coordinates preserve a nonlinear problem, but they do not solve it. Once the equation is written in packet coordinates, its derivatives, compositions, correction maps, refinement failures, and gluing defects can be separated without discarding mixed terms.

The first stage of Part II treats the local nonlinear calculus. Packet analysis and synthesis conjugate arbitrary maps. Fréchet derivatives split into multilinear packet jets. The higher chain rule then becomes a Faà di Bruno formula whose terms carry intermediate packet labels. Standard background on Fréchet calculus and the higher chain rule may be found in [1, 2, 4].

The next stage studies nonlinear iteration. Repeated substitution is indexed by rooted trees rather than operator words. A scalar majorant gives an absolute convergence theorem for polynomial fixed point equations. The quadratic case recovers the Catalan series. Rooted tree combinatorics are discussed in [16, 11, 6].

The correction stage concerns correction and obstruction. A linear correction map

$$D : U \longrightarrow V$$

places the obstruction to solving $Du = -r$ in the canonical quotient $\operatorname{coker} D$. This quotient has a universal property and behaves naturally under commuting squares. Mapping cones extend the same construction to cochain complexes. In a Hilbert space with closed range, the quotient is

represented by the orthogonal projection onto $\operatorname{Ker} D^*$. The Moore Penrose inverse then gives the least norm correction. Relevant background appears in [18, 15].

A vanishing linear obstruction is only the first step. We give a contraction theorem for nonlinear correction, followed by a local Lyapunov Schmidt reduction. In a complete filtered problem, every lifting step has its own cokernel class. Vanishing at all depths and compatible choices of corrections produce an exact inverse limit solution. A filtered correction complex carries a spectral sequence. Its first page records graded obstruction candidates, while later differentials record interactions that can cancel them. We use the standard spectral sequence of a filtered complex from [18, 12].

The final stage studies elimination and gluing. The implicit function theorem gives an exact Schur derivative after hidden variables are removed. Finite transport systems reduce, after a spanning tree gauge, to one holonomy element for each fundamental cycle. The same idea extends to smooth projector families through Kato transport. We use standard material from [4, 5, 7, 8, 10].

The article does not claim that every nonlinear problem has a finite obstruction space. The cokernel is canonical only after a correction map has been specified. It may be infinite dimensional. A first order obstruction may vanish while a higher order obstruction remains. The purpose of the calculus is to identify the exact object in which each failure lives and to show how those objects change under packets, refinement, filtration, and transport.

15 A map of the nonlinear calculus

Four different questions recur throughout the paper. They should not be merged into one vague idea of nonlinear structure.

The first question is representation. Given a nonlinear map F, can its packet form be written without approximation? The answer is yes. Analysis and synthesis are inverse coordinate maps, so conjugation preserves the full equation.

The second question is interaction. Which input packets influence a given output packet, and at what derivative order? Packet jets answer this question. They form a directed hypergraph rather than an ordinary block matrix because one nonlinear interaction can have several inputs.

The third question is correction. Suppose an approximate solution leaves a defect r. A declared linearized correction map D removes the part of r lying in $\operatorname{im} D$. The quotient class

$$[r] \in \operatorname{coker} D$$

is the part that no allowed first order correction can remove. This is the obstruction. A nonlinear theorem is still needed after that class vanishes.

The fourth question is gluing. Local coordinate changes can solve every edge problem and still fail around a loop. The remaining cycle product is holonomy. It is an obstruction to one global frame, not an obstruction to solving a single local equation.

These four layers fit into the sequence

$$\begin{aligned}\text{exact coordinates} \longrightarrow \text{packet interactions} \longrightarrow \text{correction classes} \\ \longrightarrow \text{global gluing.}\end{aligned}$$

Each arrow needs hypotheses. Exact coordinates need a complete packet system. Differential interactions need regularity. A cokernel obstruction needs a specified correction map. Holonomy needs invertible local transports.

A second distinction concerns scale. A refinement defect compares a fine nonlinear model with a coarse one. It is a point in the coarse output space. An obstruction class is the image of such a defect, or of another residual, in a quotient by allowed corrections. The two are related only after the correction map has been declared. A small defect can have a nonzero obstruction, and a large defect can have zero obstruction if it lies entirely in the correction range.

The paper uses finite packet systems in the coordinate and graph sections. Banach space hypotheses enter only where norms, contractions, or Fréchet derivatives are used. Module and complex language is used for cokernels, filtered lifting, and cohomological obstructions. This separation keeps each result at the level of generality supported by its proof.

Standing conventions

All packet systems are finite unless a statement says otherwise. In a normed setting every packet projector and every coarsening operator is bounded. Differentiability of operator valued maps is taken in operator norm. A Banach space isomorphism is a bounded linear bijection with bounded inverse. For a decreasing filtration $(F^m X)$, completeness means that the canonical map

$$X \longrightarrow \varprojlim_m X/F^m X$$

is an isomorphism and separatedness means $\bigcap_m F^m X = 0$. Statements about congruence classes in a filtered nonlinear problem also assume that the nonlinear map preserves filtration congruences at the stated depth. These assumptions are repeated where they are essential to a proof.

16 Exact nonlinear packet coordinates

Let X be a module over a unital ring. A finite packet system on X is a family

$$\mathcal{P} = (P_\alpha)_{\alpha \in I} \subseteq \mathrm{End}(X)$$

of nonzero idempotents such that

$$P_\alpha P_\beta = 0 \quad \text{when } \alpha \neq \beta, \qquad \sum_{\alpha \in I} P_\alpha = I_X.$$

Write

$$X_\alpha = P_\alpha X, \qquad X^{\mathcal{P}} = \bigoplus_{\alpha \in I} X_\alpha.$$

The analysis and synthesis maps are

$$\mathcal{A}_{\mathcal{P}} x = (P_\alpha x)_\alpha, \qquad \mathcal{S}_{\mathcal{P}} (x_\alpha)_\alpha = \sum_\alpha x_\alpha.$$

They are inverse isomorphisms.

Definition 16.1 (Nonlinear packet coordinate map)**.** Let $\mathcal{P}$ be a packet system on X and let $\mathcal{Q}$ be a packet system on Y. For any map $F : X \to Y$, define

$$F_{\mathcal{Q}\leftarrow\mathcal{P}} = \mathcal{A}_{\mathcal{Q}} F \mathcal{S}_{\mathcal{P}}.$$

Thus

$$F_{\mathcal{Q}\leftarrow\mathcal{P}}((x_\alpha)_\alpha) = \left(Q_\gamma F \left(\sum_\alpha x_\alpha \right) \right)_\gamma .$$

Theorem 16.2 (Exact nonlinear coordinate theorem)**.** *For every map $F : X \to Y$,*

$$\mathcal{S}_{\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} \mathcal{A}_{\mathcal{P}} = F.$$

If $G : Y \to Z$ and $\mathcal{R}$ is a packet system on Z, then

$$(G \circ F)_{\mathcal{R}\leftarrow\mathcal{P}} = G_{\mathcal{R}\leftarrow\mathcal{Q}} \circ F_{\mathcal{Q}\leftarrow\mathcal{P}}.$$

No linearity, continuity, or differentiability assumption is needed.

Proof. The identities $\mathcal{S}_{\mathcal{P}}\mathcal{A}_{\mathcal{P}} = I_X$ and $\mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} = I_Y$ give

$$\mathcal{S}_{\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} \mathcal{A}_{\mathcal{P}} = \mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} F \mathcal{S}_{\mathcal{P}}\mathcal{A}_{\mathcal{P}} = F.$$

For composition,

$$G_{\mathcal{R}\leftarrow\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} = \mathcal{A}_{\mathcal{R}} G \mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} F \mathcal{S}_{\mathcal{P}} = \mathcal{A}_{\mathcal{R}} (G \circ F) \mathcal{S}_{\mathcal{P}}.$$

□

Corollary 16.3 (Dynamical equivalence)**.** *Suppose $X = Y$ and the same packet system is used on both sides. Packet coordinates preserve fixed points, periodic points, injectivity, surjectivity, and invertibility. In particular,*

$$F(x) = x \quad \Longleftrightarrow \quad F_{\mathcal{P}\leftarrow\mathcal{P}}(\mathcal{A}_{\mathcal{P}} x) = \mathcal{A}_{\mathcal{P}} x.$$

Proof. The packet map is conjugate to F through the isomorphism $\mathcal{A}_{\mathcal{P}}$. Every listed property is invariant under conjugacy. □

Proposition 16.4 (Change of packet coordinates)**.** *Let $\mathcal{P}$ and $\mathcal{P}'$ be packet systems on X. Let $\mathcal{Q}$ and $\mathcal{Q}'$ be packet systems on Y. For any two packet systems $\mathcal{U}$ and $\mathcal{V}$ on the same space, write*

$$J_{\mathcal{U}\leftarrow\mathcal{V}} = \mathcal{A}_{\mathcal{U}} \mathcal{S}_{\mathcal{V}}.$$

Then

$$F_{\mathcal{Q}'\leftarrow\mathcal{P}'} = J_{\mathcal{Q}'\leftarrow\mathcal{Q}} F_{\mathcal{Q}\leftarrow\mathcal{P}} J_{\mathcal{P}\leftarrow\mathcal{P}'}.$$

Proof. Insert $\mathcal{S}_{\mathcal{Q}}\mathcal{A}_{\mathcal{Q}} = I_Y$ and $\mathcal{S}_{\mathcal{P}}\mathcal{A}_{\mathcal{P}} = I_X$ into the definition of $F_{\mathcal{Q}'\leftarrow\mathcal{P}'}$. □

Example 16.5 (A mixed quadratic map)**.** Let $X = Y = \mathbb{k}^2$ with coordinate packets. For

$$F(x_1, x_2) = (x_1 x_2, x_1^2 + x_2),$$

the first output packet depends on both input packets. The second output packet contains a quadratic contribution from the first input packet and a linear contribution from the second. Packet coordinates record these interactions exactly. Replacing F by the two diagonal restrictions would lose the product x_1x_2.

Remark 16.6. Exact packet coordinates do not imply packet independence. They retain the complete input tuple. Every output packet may depend on every input packet.

17 Packet jets and higher composition

Let X and Y be Banach spaces and let the packet projectors on both spaces be bounded. Assume that $F : X \to Y$ is m times continuously Fréchet differentiable. The derivative

$$D^rF(x) : X^r \longrightarrow Y$$

is a continuous symmetric multilinear map.

Definition 17.1 (Packet jet)**.** For $1 \le r \le m$, define

$$J^r_{\gamma|\alpha_1,\dots,\alpha_r}F(x) = Q_\gamma D^rF(x)\big|_{X_{\alpha_1}\times\cdots\times X_{\alpha_r}}.$$

This is a continuous multilinear map

$$X_{\alpha_1} \times \cdots \times X_{\alpha_r} \longrightarrow Y_\gamma.$$

Theorem 17.2 (Exact reconstruction of derivatives)**.** *For $h_1, \dots, h_r \in X$,*

$$D^rF(x)[h_1,\dots,h_r] = \sum_\gamma \sum_{\alpha_1,\dots,\alpha_r} J^r_{\gamma|\alpha_1,\dots,\alpha_r}F(x)[P_{\alpha_1}h_1,\dots,P_{\alpha_r}h_r].$$

Proof. Insert $I_Y = \sum_\gamma Q_\gamma$ on the output and $I_X = \sum_\alpha P_\alpha$ in every input slot. Multilinearity expands the finite sums. □

Proposition 17.3 (Jet norm bound)**.** *The operator norm of each packet jet satisfies*

$$\left\|J^r_{\gamma|\alpha_1,\dots,\alpha_r}F(x)\right\| \le \|Q_\gamma\|\,\|D^rF(x)\|\prod_{j=1}^r \|P_{\alpha_j}\|.$$

When all packets are orthogonal projections on Hilbert spaces, every nonzero projector has norm one and the bound reduces to the norm of the full derivative.

Proof. Apply the definition of the operator norm to the composition of $D^rF(x)$ with the input and output projectors. □

Corollary 17.4 (Packet Taylor formula)**.** *Assume F is $m+1$ times continuously Fréchet differentiable on an open set containing the segment $x+th$ for $0 \le t \le 1$. Then*

$$\begin{aligned} Q_\gamma F(x+h) = Q_\gamma F(x) &+ \sum_{r=1}^m \frac{1}{r!}\sum_{\alpha_1,\dots,\alpha_r} J^r_{\gamma|\alpha_1,\dots,\alpha_r}F(x)[P_{\alpha_1}h,\dots,P_{\alpha_r}h] \\ &+ Q_\gamma R_{m+1}(x,h), \end{aligned}$$

where the Bochner integral remainder is

$$R_{m+1}(x,h) = \frac{1}{m!}\int_0^1 (1-t)^m D^{m+1}F(x+th)[h,\dots,h]\,dt.$$

Proof. The Banach space Taylor formula with integral remainder gives the unprojected identity. Apply Q_γ to every term and use theorem 17.2 on each derivative. Boundedness of Q_γ allows it to pass through the Bochner integral. □

For a fixed base point, the nonzero packet jets form a directed hypergraph. An order r interaction has r input packet labels and one output packet label. This support depends on the chosen packet system.

For a block $B = \{i_1 < \cdots < i_s\}$ and an intermediate packet γ, write

$$G_{\gamma,B}(x \mid h) = J^s_{\gamma|\alpha_{i_1},\dots,\alpha_{i_s}} G(x)[h_{i_1},\dots,h_{i_s}].$$

Theorem 17.5 (Packet Faà di Bruno formula)**.** *Let $G : X \to Y$ and $F : Y \to Z$ be n times continuously Fréchet differentiable. Let $\mathcal{P}$, $\mathcal{Q}$, and $\mathcal{R}$ be packet systems on X, Y, and Z. Fix input packets $\alpha_1,\dots,\alpha_n$, an output packet δ, and directions $h_i \in X_{\alpha_i}$. Then*

$$\begin{aligned} &J^n_{\delta|\alpha_1,\dots,\alpha_n}(F\circ G)(x)[h_1,\dots,h_n] \\ &= \sum_{\pi=\{B_1,\dots,B_k\}} \sum_{\gamma_1,\dots,\gamma_k} J^k_{\delta|\gamma_1,\dots,\gamma_k} F(Gx)\big[G_{\gamma_1,B_1}(x \mid h),\dots,G_{\gamma_k,B_k}(x \mid h)\big]. \end{aligned}$$

The first sum runs over all set partitions of $\{1,\dots,n\}$.

Proof. The higher chain rule writes $D^n(F\circ G)(x)$ as a sum over set partitions. Insert $I_Y = \sum_\gamma Q_\gamma$ in every argument of the derivative of F. Insert the input and output packet identities into each derivative of G and into the final output. Group the terms by the partition and by the intermediate packet labels. □

Proposition 17.6 (Number of labeled composition terms)**.** *If Y has M packets, then the number of packet labeled terms at order n before zero jets are removed is*

$$N_{n,M} = \sum_{k=1}^{n} S(n,k) M^k,$$

where $S(n,k)$ is a Stirling number of the second kind. Thus $N_{n,M}$ is the nth Touchard polynomial evaluated at M.

Proof. A set partition with k blocks receives one independent intermediate packet label on each block. It therefore gives M^k labeled terms. There are $S(n,k)$ such partitions. □

Corollary 17.7 (Structural pruning)**.** *A term in theorem 17.5 vanishes whenever one of its constituent packet jets vanishes. The support of the composed jet is therefore contained in the set of decorated partitions built from the supports of the jets of F and G.*

Proof. Every summand is a composition of the displayed packet jets. A zero factor makes the entire summand zero. □

Example 17.8 (The second order composition law)**.** At order two, the partition formula has two shapes. For packet directions h_1 and h_2,

$$Q_\delta D^2(F\circ G)(x)[h_1,h_2] = \sum_{\gamma_1,\gamma_2} J^2_{\delta|\gamma_1,\gamma_2}F(Gx)[J^1_{\gamma_1|\alpha_1}G(x)h_1, J^1_{\gamma_2|\alpha_2}G(x)h_2] + \sum_{\gamma} J^1_{\delta|\gamma}F(Gx)[J^2_{\gamma|\alpha_1,\alpha_2}G(x)[h_1,h_2]].$$

The first line records two first order changes entering a quadratic interaction of F. The second records a genuine second order interaction created inside G before F is applied.

Remark 17.9. Jet sparsity is not invariant under an arbitrary change of coordinates. A useful packet system normally comes from symmetry, scale, support, grading, valuation, or another structure that is already present in the problem. Every sparsity claim should therefore name the packet system with respect to which it is made.

18 Rooted tree expansions

Repeated nonlinear substitution is indexed by rooted trees. We begin with a polynomial fixed point equation on a Banach space.

Let X be a Banach space. Fix $s\in X$ and continuous multilinear maps

$$B_r : X^r \longrightarrow X, \qquad 2\le r\le d.$$

Set

$$a=\|s\|, \qquad b_r=\|B_r\|, \qquad \phi(t)=a+\sum_{r=2}^{d} b_r t^r.$$

An ordered rooted tree has leaves labeled by s and every internal vertex of arity r labeled by B_r. Its value is obtained by evaluating from the leaves toward the root.

Theorem 18.1 (Polynomial tree expansion)**.** *Suppose there is $R>0$ such that*

$$\phi(R)\le R$$

and

$$q=\sum_{r=2}^{d} r b_r R^{r-1} < 1.$$

Then the map

$$\Phi(x)=s+\sum_{r=2}^{d} B_r(x,\dots,x)$$

has a unique fixed point in the closed ball $\overline{B}_R(0)$. The sum of the values of all finite ordered rooted trees converges absolutely to that fixed point.

Proof. For $x,y\in\overline{B}_R(0)$, multilinearity and telescoping in the r input positions give

$$\|B_r(x,\dots,x)-B_r(y,\dots,y)\|\le r b_r R^{r-1}\|x-y\|.$$

Hence Φ is q Lipschitz on the ball. The inequality $\phi(R) \le R$ shows that Φ preserves the ball. The contraction mapping theorem gives a unique fixed point u in that ball.

Let $\mathfrak{T}_m$ be the finite set of ordered rooted trees of height less than m. A leaf has value s. If the root of τ has arity r and ordered subtrees $\tau_1, \dots, \tau_r$, define

$$B_\tau(s) = B_r(B_{\tau_1}(s), \dots, B_{\tau_r}(s)).$$

Induction on m shows that the Picard iterate $u_0 = 0$ and $u_{m+1} = \Phi(u_m)$ satisfies

$$u_m = \sum_{\tau \in \mathfrak{T}_m} B_\tau(s).$$

Define a nonnegative scalar weight by assigning weight a to a leaf and applying the scalar map $(t_1, \dots, t_r) \mapsto b_r t_1 \cdots t_r$ at a vertex of arity r. A second induction gives

$$\sum_{\tau \in \mathfrak{T}_m} \|B_\tau(s)\| \le \sum_{\tau \in \mathfrak{T}_m} w(\tau) = t_m,$$

where $t_0 = 0$ and $t_{m+1} = \phi(t_m)$. The sequence (t_m) is increasing because ϕ has nonnegative coefficients. It is bounded by R because $t_0 \le R$ and $\phi(R) \le R$. The sum of the norms of all tree values is therefore finite. The tree series converges absolutely. Its partial sums are the Picard iterates, so its sum is u. □

Corollary 18.2 (Quadratic Catalan expansion)**.** *Let $B : X \times X \to X$ be bounded and consider*

$$u = s + B(u, u).$$

Put $a = \|s\|$ and $b = \|B\|$. If $4ab < 1$, then

$$u = \sum_{n \ge 1} \sum_{\substack{\tau \text{ full ordered binary tree} \\ \text{with } n \text{ leaves}}} B_\tau(s)$$

converges absolutely. It also satisfies

$$\|u\| \le \frac{1 - \sqrt{1 - 4ab}}{2b}$$

when $b > 0$.

Proof. If $b = 0$, the equation is $u = s$ and the tree series contains only its one leaf. If $a = 0$, every tree value is zero. Choose any R with $0 < R < 1/(2b)$ and apply theorem 18.1 to obtain uniqueness of the zero fixed point in that ball.

Assume now that $a > 0$ and $b > 0$. Take

$$R = \frac{1 - \sqrt{1 - 4ab}}{2b}.$$

Then $a + bR^2 = R$ and $2bR < 1$. Apply theorem 18.1. The number of full ordered binary trees with n leaves is the Catalan number C_{n-1}, which gives the displayed majorant. □

Theorem 18.3 (Packet labeled tree reconstruction)**.** *Let X carry a packet system $\mathcal{P}$. At each internal vertex of a tree, insert the output identity and one input identity for every child. Summing over all internal packet labels reconstructs the original tree value exactly.*

Proof. At a vertex labeled by an r multilinear map B_r, insert

$$I_X = \sum_{\beta} P_\beta$$

on the output and in every input slot. Multilinearity gives an exact finite sum of packet blocks at that vertex. Apply this identity successively from the leaves toward the root. Since the tree and the packet set are finite, all sums are finite and their total reconstructs the original tree value. □

For an infinite tree expansion, each underlying tree's finite sum of packet decorations is kept grouped. The unlabelled tree majorant does not imply absolute convergence of the fully flattened family of decorations for arbitrary oblique projectors. Finite tree reconstruction and this grouped interpretation remain exact.

Corollary 18.4 (Tree pruning)**.** *A packet labeled tree vanishes if one internal vertex carries a zero packet block of the corresponding multilinear map. Removing all such zero terms leaves every finite tree sum unchanged and therefore leaves the grouped infinite series unchanged.*

Proof. A zero vertex map makes the value of the full tree zero. □

The tree theorem separates combinatorial growth from analytic convergence. The number of trees may grow rapidly, while the norm of each tree decreases through the multilinear bounds. The scalar map ϕ is the exact majorant that balances these two effects. Packet pruning changes the combinatorics but not the proof. It removes trees whose values are already zero before any norm estimate is used.

Example 18.5 (Catalan parity packets)**.** Over a field of characteristic zero, the formal equation

$$u = z + u^2$$

has the branch

$$u(z) = \frac{1 - \sqrt{1 - 4z}}{2}.$$

Write $u = u_0 + u_1$, where u_0 contains the even powers of z and u_1 contains the odd powers. Exact packet projection gives

$$u_0 = u_0^2 + u_1^2, \qquad u_1 = z + 2u_0u_1.$$

The two equations remain coupled. Their sum is the original fixed point equation.

19 Graded nonlinear operations

Let G be a finite abelian group and let

$$A = \bigoplus_{g \in G} A_g$$

be a G graded module.

Definition 19.1 (Grade respecting operation)**.** An r multilinear map $\mu : A^r \to A$ respects the grading when

$$\mu(A_{g_1}, \dots, A_{g_r}) \subseteq A_{g_1 + \cdots + g_r}.$$

Theorem 19.2 (Graded packet convolution)**.** *If $a_j = \sum_{g \in G} a_{j,g}$ and μ respects the grading, then*

$$P_k \mu(a_1, \dots, a_r) = \sum_{g_1 + \cdots + g_r = k} \mu(a_{1,g_1}, \dots, a_{r,g_r}).$$

Proof. Expand every input into its homogeneous components. Each term lands in the component whose label is the sum of its input labels. Projection onto A_k retains exactly the displayed terms. □

Corollary 19.3 (Cyclic product formula)**.** *For a cyclically graded algebra*

$$A = \bigoplus_{j \in \mathbb{Z}/N\mathbb{Z}} A_j, \qquad A_i A_j \subseteq A_{i+j},$$

one has

$$P_k(ab) = \sum_{i+j=k} a_i b_j.$$

Proof. Apply theorem 19.2 to multiplication. □

Theorem 19.4 (Packet Maurer Cartan equations)**.** *Let*

$$\mathcal{L} = \bigoplus_{g \in G} \mathcal{L}_g$$

be a differential graded Lie algebra over a ring in which two is invertible. Assume

$$d\mathcal{L}_g \subseteq \mathcal{L}_g, \qquad [\mathcal{L}_g, \mathcal{L}_h] \subseteq \mathcal{L}_{g+h}.$$

For a degree one element $\eta = \sum_g \eta_g$, the Maurer Cartan equation

$$d\eta + \frac{1}{2}[\eta, \eta] = 0$$

is equivalent to

$$d\eta_k + \frac{1}{2} \sum_{g+h=k} [\eta_g, \eta_h] = 0$$

for every $k \in G$.

Proof. The differential preserves the group label and the bracket adds labels. Apply the projector onto $\mathcal{L}_k$ and use exact reconstruction. □

The same argument applies to higher brackets.

Proposition 19.5 (Packet equations for L_∞ operations)**.** *Let $\mathcal{L}$ be complete and carry operations ℓ_r that respect a finite abelian group grading. Assume the packet projectors are continuous for the topology in which the series below converges. Assume the Maurer Cartan series*

$$\sum_{r\geq 1}\frac{1}{r!}\ell_r(\eta,\dots,\eta)$$

converges and the required factorials are invertible. Then its packet of label k is

$$\sum_{r\geq 1}\frac{1}{r!}\sum_{g_1+\cdots+g_r=k}\ell_r(\eta_{g_1},\dots,\eta_{g_r}).$$

The full Maurer Cartan equation holds if and only if every packet equation vanishes.

Proof. Apply theorem 19.2 to each operation ℓ_r. Continuity of the packet projectors permits the convergent sums to be projected term by term. □

Definition 19.6 (Divided square)**.** A divided square on degree one elements is a map q satisfying

$$q(x+y)=q(x)+q(y)+[x,y],\qquad q(ax)=a^2q(x),$$

whenever the expressions are defined.

Proposition 19.7 (Divided square packet equations)**.** *Assume $q(\mathcal{L}_g)\subseteq\mathcal{L}_{2g}$. Choose an ordering of the finite label set. The equation*

$$d\eta+q(\eta)=0$$

is equivalent to

$$d\eta_k+\sum_{2g=k}q(\eta_g)+\sum_{\substack{g<h\\ g+h=k}}[\eta_g,\eta_h]=0$$

for every k.

Proof. Repeated use of the polarization identity for q gives

$$q\left(\sum_g\eta_g\right)=\sum_g q(\eta_g)+\sum_{g<h}[\eta_g,\eta_h].$$

Project onto the component of label k. □

Example 19.8 (Matrix difference grading)**.** Let $A=M_N(\Bbbk)$ and assign the matrix unit E_{rs} the label $r-s$ modulo N. Since

$$E_{rs}E_{uv}=\delta_{su}E_{rv},$$

the output label is

$$(r-s)+(u-v)=r-v$$

whenever the product is nonzero. Matrix multiplication therefore obeys the cyclic convolution law exactly. The same grading resolves commutators and polynomial matrix equations.

Remark 19.9. These results begin with a graded differential or higher algebra. They do not assert that every deformation problem has such a controlling object. Differential graded and infinity algebra methods are developed in [11].

20 Nonlinear refinement defects

Let

$$F_f : X_f \longrightarrow Y_f, \qquad F_c : X_c \longrightarrow Y_c$$

be fine and coarse maps. Let

$$C_X : X_f \longrightarrow X_c, \qquad C_Y : Y_f \longrightarrow Y_c$$

be linear coarsening maps. Whenever a norm, a Lipschitz constant, or a Fréchet derivative is used, the spaces are Banach spaces and the coarsening maps are bounded.

Definition 20.1 (Nonlinear refinement defect)**.** The pointwise defect of F is

$$\Delta_F(x) = C_Y F_f(x) - F_c(C_X x).$$

The two models are exactly compatible when Δ_F vanishes everywhere.

The defect is an ordinary vector in Y_c. It becomes an obstruction only after a correction map into Y_c is chosen and the defect is passed to the corresponding cokernel. This distinction matters in applications. Norm estimates measure the size of the defect, while a quotient determines whether the defect is removable.

Theorem 20.2 (Exact defect under composition)**.** *Let $G_f : Y_f \to Z_f$ and $G_c : Y_c \to Z_c$, with coarsening $C_Z : Z_f \to Z_c$. Then*

$$\Delta_{G\circ F}(x) = \Delta_G(F_f(x)) + G_c(F_c(C_X x) + \Delta_F(x)) - G_c(F_c(C_X x)).$$

If G_c is L Lipschitz on the relevant set, then

$$\|\Delta_{G\circ F}(x)\| \le \|\Delta_G(F_f(x))\| + L\|\Delta_F(x)\|.$$

Proof. Add and subtract $G_c(C_Y F_f(x))$. Use

$$C_Y F_f(x) = F_c(C_X x) + \Delta_F(x).$$

The norm estimate follows from the Lipschitz bound. □

Corollary 20.3 (Linear product rule)**.** *If all maps in theorem 20.2 are linear, then*

$$\Delta_{GF} = \Delta_G F_f + G_c \Delta_F.$$

Proof. Linearity turns the last difference in theorem 20.2 into $G_c\Delta_F(x)$. □

Theorem 20.4 (Iterated orbit defect)**.** *Let $F_f : X_f \to X_f$ and $F_c : X_c \to X_c$, with coarsening $C : X_f \to X_c$. Suppose F_c is q Lipschitz. For a fine orbit starting at x, assume*

$$\|\Delta_F(F_f^j x)\| \le \varepsilon_j.$$

Then

$$\left\|CF_f^n(x) - F_c^n(Cx)\right\| \le \sum_{j=0}^{n-1} q^{n-1-j}\varepsilon_j.$$

If $0 \le q < 1$ and every $\varepsilon_j \le \varepsilon$, then

$$\left\|CF_f^n(x) - F_c^n(Cx)\right\| \le \varepsilon\frac{1-q^n}{1-q}.$$

Proof. Let

$$e_n = \left\|CF_f^n(x) - F_c^n(Cx)\right\|.$$

The definition of the defect and the Lipschitz property give

$$e_{n+1} \le \varepsilon_n + qe_n.$$

Iterating this recurrence proves the first estimate. The second is the geometric sum. □

Theorem 20.5 (Fixed point stability)**.** *Assume F_c is a contraction with constant $q < 1$. Let x_f and x_c be fixed points of F_f and F_c. If*

$$\|\Delta_F(x_f)\| \le \varepsilon,$$

then

$$\|Cx_f - x_c\| \le \frac{\varepsilon}{1-q}.$$

When the defect at x_f is zero, the fixed points are compatible.

Proof. Using the fixed point equations,

$$\begin{aligned}\|Cx_f - x_c\| &\le \|CF_f(x_f) - F_c(Cx_f)\| + \|F_c(Cx_f) - F_c(x_c)\| \\ &\le \varepsilon + q\|Cx_f - x_c\|.\end{aligned}$$

Rearrange. □

Corollary 20.6 (Several refinement levels)**.** *Let $C_\ell : X_{\ell+1} \to X_\ell$ for $0 \le \ell < m$. Suppose x_ℓ is a fixed point at level ℓ, the coarse map at level ℓ has contraction constant $q_\ell < 1$, and the adjacent defect at $x_{\ell+1}$ is at most ε_ℓ. Write*

$$C_{a,b} = C_aC_{a+1}\cdots C_{b-1} \quad \text{for } a < b, \qquad C_{a,a} = I.$$

Then

$$\|C_{0,m}x_m - x_0\| \le \sum_{\ell=0}^{m-1} \|C_{0,\ell}\|\frac{\varepsilon_\ell}{1-q_\ell}.$$

Proof. Insert the intermediate vectors $C_{0,\ell}x_\ell$ and telescope. Apply theorem 20.5 to every adjacent pair. □

Proposition 20.7 (Derivative defects)**.** *Assume F_f and F_c are r times Fréchet differentiable and the coarsening maps are bounded and linear. Then*

$$D\Delta_F(x) = C_Y DF_f(x) - DF_c(C_X x)C_X.$$

For $r \geq 2$,

$$D^r\Delta_F(x)[h_1, \ldots, h_r] = C_Y D^r F_f(x)[h_1, \ldots, h_r] - D^r F_c(C_X x)[C_X h_1, \ldots, C_X h_r].$$

Proof. Differentiate the defining identity for Δ_F. The coarsening maps are linear, so their higher derivatives vanish. □

Example 20.8 (Sharp affine bound)**.** Let $F_c(y) = qy + b$ with $0 < q < 1$. Choose a fine fixed point whose coarsening has constant defect ε in the direction of the scalar state. The fixed point mismatch is exactly $\varepsilon/(1-q)$. Thus the constant in theorem 20.5 cannot be improved without more information.

21 Canonical obstruction objects

We begin by isolating the part of a defect that cannot be removed by an allowed linear correction.

Definition 21.1 (Linear correction datum)**.** A linear correction datum consists of a homomorphism

$$D : U \longrightarrow V$$

and a defect $r \in V$. A correction is an element $u \in U$ satisfying

$$Du = -r.$$

The obstruction class is

$$\mathrm{Ob}_D(r) = [r] \in \operatorname{coker} D = V/\operatorname{im} D.$$

Theorem 21.2 (Cokernel obstruction criterion)**.** *A linear correction exists if and only if*

$$\mathrm{Ob}_D(r) = 0.$$

The quotient map

$$q_D : V \longrightarrow \operatorname{coker} D$$

has the following universal property. If $\varphi : V \to W$ is linear and $\varphi D = 0$, then there is a unique linear map

$$\overline{\varphi} : \operatorname{coker} D \longrightarrow W$$

with $\varphi = \overline{\varphi} q_D$.

Proof. The class of r is zero exactly when $r \in \operatorname{im} D$, which is equivalent to solvability of $Du = -r$. If $\varphi D = 0$, then φ vanishes on $\operatorname{im} D$. It therefore descends uniquely to the quotient. □

The universal property explains why the cokernel is the canonical obstruction object relative to D. Any linear obstruction test that vanishes on correctable defects factors through it.

Theorem 21.3 (Naturality of obstruction classes)**.** *Suppose*

$$\begin{array}{ccc} U & \xrightarrow{D} & V \\ a\downarrow & & \downarrow b \\ U' & \xrightarrow{D'} & V' \end{array}$$

is a commuting square. Then there is a unique induced map

$$\overline{b} : \operatorname{coker} D \longrightarrow \operatorname{coker} D'$$

satisfying

$$\overline{b}\,\operatorname{Ob}_D(r) = \operatorname{Ob}_{D'}(br).$$

Proof. The relation $bD = D'a$ shows that the quotient map $q_{D'}b$ vanishes on $\operatorname{im} D$. Apply the universal property in theorem 21.2. □

Theorem 21.4 (Packet decomposition of an obstruction)**.** *Suppose*

$$U = \bigoplus_{\alpha\in I} U_\alpha, \qquad V = \bigoplus_{\alpha\in I} V_\alpha,$$

and $D(U_\alpha) \subseteq V_\alpha$ for every α. Let

$$D_\alpha : U_\alpha \longrightarrow V_\alpha$$

be the restriction. Then

$$\operatorname{coker} D \cong \bigoplus_{\alpha\in I} \operatorname{coker} D_\alpha.$$

Under this isomorphism, the obstruction class of $r = \sum_\alpha r_\alpha$ is the tuple

$$\left([r_\alpha]\right)_\alpha.$$

Hence r is correctable if and only if every packet obstruction vanishes.

Proof. The grading assumption gives

$$\operatorname{im} D = \bigoplus_{\alpha} \operatorname{im} D_\alpha.$$

Taking the quotient of the direct sum by this direct sum of images gives the displayed isomorphism. □

Remark 21.5. The conclusion of theorem 21.4 requires the correction map to preserve the packets. Off diagonal correction blocks can couple packet defects. In that case the full cokernel remains canonical, but it need not split into the separate packet cokernels.

Proposition 21.6 (Projectors on the obstruction quotient)**.** *Let $(Q_\alpha)_{\alpha\in I}$ be a complete family of pairwise annihilating idempotents on V. Assume*

$$Q_\alpha(\operatorname{im} D) \subseteq \operatorname{im} D$$

for every α. Then each Q_α induces an idempotent $\overline{Q}_\alpha$ on $\operatorname{coker} D$*, and*

$$\operatorname{coker} D = \bigoplus_{\alpha \in I} \overline{Q}_\alpha(\operatorname{coker} D).$$

For every defect $r \in V$,

$$\overline{Q}_\alpha \operatorname{Ob}_D(r) = [Q_\alpha r].$$

No packet decomposition of the correction space U is required.

Proof. If $v - v' \in \operatorname{im} D$, then

$$Q_\alpha(v - v') \in \operatorname{im} D.$$

Hence $[v] \mapsto [Q_\alpha v]$ is well defined on the quotient. The idempotent, annihilation, and reconstruction identities descend from V to $V/\operatorname{im} D$. The formula for an obstruction class follows from the definition. □

21.1 Relative obstruction under coarsening

A coarse problem can forget an obstruction that is still present in the fine problem. The next exact sequence identifies the part that is lost.

Theorem 21.7 (Relative obstruction exact sequence)**.** *Suppose the following diagram of modules commutes and has exact rows:*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & U_0 & \xrightarrow{i_U} & U_f & \xrightarrow{p_U} & U_c & \longrightarrow & 0 \\ & & \downarrow{\scriptstyle D_0} & & \downarrow{\scriptstyle D_f} & & \downarrow{\scriptstyle D_c} & & \\ 0 & \longrightarrow & V_0 & \xrightarrow{i_V} & V_f & \xrightarrow{p_V} & V_c & \longrightarrow & 0. \end{array}$$

There is a natural exact sequence

$$\begin{aligned} 0 \longrightarrow \operatorname{Ker} D_0 \longrightarrow \operatorname{Ker} D_f \longrightarrow \operatorname{Ker} D_c \xrightarrow{\delta} \operatorname{coker} D_0 \\ \longrightarrow \operatorname{coker} D_f \longrightarrow \operatorname{coker} D_c \longrightarrow 0. \end{aligned}$$

For $u_c \in \operatorname{Ker} D_c$, choose $u_f \in U_f$ with $p_U u_f = u_c$. There is a unique $v_0 \in V_0$ such that

$$D_f u_f = i_V v_0,$$

and the connecting map is

$$\delta(u_c) = [v_0].$$

Proof. Since $D_c u_c = 0$, commutativity gives

$$p_V D_f u_f = D_c p_U u_f = 0.$$

Exactness of the lower row gives a unique v_0 with $D_f u_f = i_V v_0$. A different lift has the form $u_f + i_U u_0$. The corresponding element of V_0 changes by $D_0 u_0$, so $[v_0]$ is independent of the lift.

The map $\operatorname{Ker} D_0 \to \operatorname{Ker} D_f$ is injective because i_U is injective. If $u_f \in \operatorname{Ker} D_f$ maps to zero in U_c, then $u_f = i_U u_0$ for a unique u_0. The identity

$$i_V D_0 u_0 = D_f i_U u_0 = 0$$

and injectivity of i_V show that $u_0 \in \operatorname{Ker} D_0$. This proves exactness at $\operatorname{Ker} D_f$.

If u_c comes from an element of $\operatorname{Ker} D_f$, one may choose that element as its lift and obtain $\delta(u_c) = 0$. Conversely, if $\delta(u_c) = 0$, then $v_0 = D_0 u_0$ for some u_0. Replacing u_f by $u_f - i_U u_0$ gives an element of $\operatorname{Ker} D_f$ that maps to u_c. This proves exactness at $\operatorname{Ker} D_c$.

The class $[v_0] \in \operatorname{coker} D_0$ maps to zero in $\operatorname{coker} D_f$ whenever it equals $\delta(u_c)$, since the chosen lift satisfies $i_V v_0 = D_f u_f$. Conversely, if $i_V v_0 = D_f u_f$, then $u_c = p_U u_f$ lies in $\operatorname{Ker} D_c$ and $\delta(u_c) = [v_0]$. This proves exactness at $\operatorname{coker} D_0$.

If a class in $\operatorname{coker} D_f$ comes from $\operatorname{coker} D_0$, its image in $\operatorname{coker} D_c$ is zero. Conversely, suppose $[v_f]$ maps to zero in $\operatorname{coker} D_c$. Choose u_c with $D_c u_c = p_V v_f$ and lift it to $u_f \in U_f$. Then

$$p_V(v_f - D_f u_f) = 0,$$

so $v_f - D_f u_f = i_V v_0$ for some $v_0 \in V_0$. Thus $[v_f]$ comes from $[v_0]$. Surjectivity at the last term follows from surjectivity of p_V. □

Corollary 21.8 (Coarse vanishing and fine vanishing)**.** *Let $[r_f] \in \operatorname{coker} D_f$. If its image in $\operatorname{coker} D_c$ is zero, then $[r_f]$ comes from $\operatorname{coker} D_0$. Coarse vanishing therefore means that the fine obstruction is supported in information removed by coarsening. It does not imply $[r_f] = 0$.*

Proof. This is exactness of the sequence in theorem 21.7 at $\operatorname{coker} D_f$. □

Example 21.9 (A fine obstruction erased by coarsening)**.** Take all three correction spaces to be zero. Let

$$V_f = \mathbb{k}^2, \qquad V_c = \mathbb{k}, \qquad p_V(x, y) = x + y,$$

and let

$$V_0 = \operatorname{Ker} p_V = \operatorname{span}\{(1, -1)\}.$$

The fine defect $(1, -1)$ is a nonzero class in $\operatorname{coker} D_f = V_f$, while its coarse image is zero. The class comes from $\operatorname{coker} D_0 = V_0$, exactly as corollary 21.8 predicts.

The same construction has a cohomological form.

Definition 21.10 (Mapping cone)**.** Let $D : C^\bullet \to E^\bullet$ be a cochain map. Its mapping cone is

$$\operatorname{Cone}(D)^n = E^n \oplus C^{n+1}$$

with differential

$$d_{\operatorname{Cone}}(e, c) = (d_E e + Dc, -d_C c).$$

Theorem 21.11 (Cohomological obstruction through the cone)**.** *Let $e \in E^n$ satisfy $d_E e = 0$. The class of $(e, 0)$ in*

$$H^n(\operatorname{Cone}(D))$$

vanishes if and only if the class $[e] \in H^n(E)$ *lies in the image of*

$$H^n(D) : H^n(C) \longrightarrow H^n(E).$$

When C *and* E *are concentrated in degree zero,*

$$H^0(\mathrm{Cone}(D)) \cong \operatorname{coker} D.$$

Proof. The class of $(e, 0)$ vanishes exactly when there are $e' \in E^{n-1}$ and $c \in C^n$ with

$$(e, 0) = d_{\mathrm{Cone}}(e', c).$$

This means

$$e = d_E e' + Dc, \qquad d_C c = 0.$$

Equivalently, $[e] = H^n(D)[c]$. The degree zero statement follows directly from the definition of the cone differential. □

Corollary 21.12 (Packet obstruction complexes)**.** *If* $C^\bullet$ *and* $E^\bullet$ *split into finite packet subcomplexes and* D *preserves those packets, then* $\mathrm{Cone}(D)$ *splits into packet subcomplexes and*

$$H^n(\mathrm{Cone}(D)) \cong \bigoplus_\alpha H^n(\mathrm{Cone}(D_\alpha)).$$

Proof. Both the cone and its cohomology commute with finite direct sums. □

Example 21.13 (A two component obstruction)**.** Let

$$D : \Bbbk \longrightarrow \Bbbk^2, \qquad D(t) = (t, t).$$

The cokernel is one dimensional. The defect (a, b) is correctable exactly when $a = b$. The obstruction can be represented by $a - b$, but this scalar representative depends on a choice of isomorphism from the cokernel to $\Bbbk$. The quotient class itself is canonical.

22 Split Hilbert obstructions and nonlinear correction

The quotient obstruction has a concrete orthogonal representative when the correction map acts between Hilbert spaces and has closed range.

Theorem 22.1 (Orthogonal obstruction projector)**.** *Let* $D : H_1 \to H_2$ *be a bounded operator between Hilbert spaces with closed range. Let* $D^\dagger$ *be its Moore Penrose inverse. Then*

$$P_{\mathrm{ob}} = I_{H_2} - DD^\dagger$$

is the orthogonal projector onto

$$(\operatorname{im} D)^\perp = \operatorname{Ker} D^*.$$

For $r \in H_2$*, the following are equivalent:*

1. $Du = -r$ *is solvable*

2. $[r] = 0$ *in* $\operatorname{coker} D$
3. $P_{\mathrm{ob}} r = 0$.

When these conditions hold, $-D^{\dagger} r$ *is the least norm correction. For every* r*,*

$$\operatorname{dist}(r, \operatorname{im} D) = \|P_{\mathrm{ob}} r\|.$$

Proof. Closed range gives the orthogonal decomposition

$$H_2 = \operatorname{im} D \oplus \operatorname{Ker} D^{*}.$$

The Moore Penrose identities show that $DD^{\dagger}$ is the orthogonal projector onto $\operatorname{im} D$. The stated equivalences and distance formula follow from this decomposition. The least norm property is the standard characterization of $D^{\dagger} r$. □

A vanishing linear obstruction does not by itself solve a nonlinear equation. The next theorem gives a quantitative condition that does.

Theorem 22.2 (Right inverse correction theorem)**.** *Let* $D : X \to Y$ *be a bounded linear map between Banach spaces. Assume* $Z = \operatorname{im} D$ *is closed and there is a bounded linear right inverse*

$$R : Z \longrightarrow X, \qquad DR = I_Z.$$

Put $X_R = R(Z)$*. Let*

$$N : \overline{B}_{\rho}(0) \cap X_R \longrightarrow Z$$

be L *Lipschitz. Let* $y \in Z$ *and put*

$$q = \|R\| L.$$

Assume $q < 1$ *and*

$$\|R(y - N(0))\| \leq (1 - q)\rho.$$

Then the equation

$$Dx + N(x) = y$$

has a unique solution in $\overline{B}_{\rho}(0) \cap X_R$*.*

Proof. The space X_R is closed. Indeed, if $Rz_n \to x$, then $z_n = DRz_n \to Dx$, so $x = R(Dx)$. Define

$$\Psi(x) = R(y - N(x)).$$

This map takes X_R to itself. For $x \in \overline{B}_{\rho}(0) \cap X_R$,

$$\begin{aligned} \|\Psi(x)\| &\leq \|R(y - N(0))\| + \|R\| \, \|N(x) - N(0)\| \\ &\leq (1 - q)\rho + q\rho = \rho. \end{aligned}$$

Also,

$$\|\Psi(x) - \Psi(x')\| \leq q \|x - x'\|.$$

The contraction mapping theorem gives a unique fixed point in the stated set. Applying D to the fixed point identity gives the equation. Conversely, a solution in X_R equals $R(Dx)$ and is therefore a fixed point of Ψ. □

Corollary 22.3 (Stability with respect to the target)**.** *Under the hypotheses of theorem 22.2, let x and x' be the fixed points associated with targets y and y'. If both fixed point maps preserve the same ball, then*

$$\|x - x'\| \leq \frac{\|R\|}{1-q}\|y - y'\|.$$

Proof. Subtract the two fixed point equations and use the Lipschitz bound for N:

$$\|x - x'\| \leq \|R\|\|y - y'\| + q\|x - x'\|.$$

□

The range equation can be separated from the remaining nonlinear obstruction. Standard forms of this reduction appear in [9, 19].

Theorem 22.4 (Local Lyapunov Schmidt reduction)**.** *Let $\mathcal{U} \subseteq X$ be an open neighborhood of zero and let $F : \mathcal{U} \to Y$ be a C^r map between Banach spaces with $r \geq 1$, $F(0) = 0$, and $L = DF(0)$. Assume there are closed decompositions*

$$X = K \oplus X_1, \qquad Y = R \oplus O,$$

where $K = \operatorname{Ker} L$, $R = \operatorname{im} L$, and

$$L_1 = L|_{X_1} : X_1 \longrightarrow R$$

is a Banach space isomorphism. Let π_R and π_O be the two projections on Y. Then there are neighborhoods $U_K \subseteq K$ and $U_1 \subseteq X_1$ of zero and a unique C^r map

$$\psi : U_K \longrightarrow U_1$$

with $\psi(0) = 0$ such that

$$\pi_R F(k + \psi(k)) = 0$$

for every $k \in U_K$. After the neighborhoods are made smaller if needed, the solutions of $F(x) = 0$ near zero are exactly the points

$$x = k + \psi(k)$$

for which the reduced obstruction map

$$\kappa(k) = \pi_O F(k + \psi(k))$$

vanishes. At the base point,

$$D\psi(0) = 0, \qquad D\kappa(0) = 0.$$

Proof. Define

$$G(k, x_1) = \pi_R F(k + x_1).$$

The derivative of G in the X_1 variable at $(0, 0)$ is L_1, which is invertible. The implicit function theorem gives the unique map ψ. Every nearby x has a unique decomposition $x = k + x_1$. The range equation forces $x_1 = \psi(k)$, and the remaining equation is precisely $\kappa(k) = 0$.

Differentiate $G(k, \psi(k)) = 0$ at zero. Since L vanishes on K, one obtains $L_1 D\psi(0) = 0$, hence $D\psi(0) = 0$. Finally,

$$D\kappa(0) = \pi_O L|_K = 0.$$

□

Corollary 22.5 (Quadratic reduced obstruction)**.** *Assume the hypotheses of theorem 22.4 with $r \geq 2$. For $k_1, k_2 \in K$,*

$$D^2\kappa(0)[k_1, k_2] = \pi_O D^2 F(0)[k_1, k_2].$$

Proof. Differentiate $\kappa(k) = \pi_O F(k + \psi(k))$ twice. The terms containing $D\psi(0)$ vanish. The term containing $D^2\psi(0)$ is multiplied by $\pi_O L$, which is zero because $L(X) = R$. □

Example 22.6 (A purely quadratic obstruction)**.** Let

$$F(x, y) = (y, x^2)$$

on $\mathbb{R}^2$. At the origin,

$$L(h, k) = (k, 0).$$

The kernel is the x axis, the range is the first output axis, and the cokernel is the second output axis. The range equation forces $y = 0$. The reduced obstruction map is

$$\kappa(x) = x^2.$$

The linear obstruction vanishes for every kernel direction, but the nonlinear equation still has only the solution $x = 0$ near the origin.

Remark 22.7. The reduced map κ begins at quadratic order. This is why vanishing of the first order cokernel class need not settle a nonlinear problem. Higher terms can produce a genuine obstruction on the kernel directions.

23 Filtered lifting

Let X and Y be modules with decreasing filtrations

$$X = F^0X \supseteq F^1X \supseteq \cdots, \qquad Y = F^0Y \supseteq F^1Y \supseteq \cdots.$$

Assume both are separated and complete for their filtration topologies. Let $\Phi : X \to Y$ be continuous and filtration congruence preserving in the sense that

$$h \in F^mX \quad \Longrightarrow \quad \Phi(x + h) - \Phi(x) \in F^mY$$

for every $x \in X$ and every m. This condition makes the statement $\Phi(x) \in F^mY$ depend only on the class of x modulo F^mX.

Suppose x_m satisfies

$$\Phi(x_m) \in F^mY.$$

Assume that corrections $h \in F^m X$ obey a first order congruence

$$\Phi(x_m + h) \equiv \Phi(x_m) + L_m h \pmod{F^{m+1}Y},$$

where $L_m : F^m X \to F^m Y$ is a module homomorphism satisfying

$$L_m(F^{m+1}X) \subseteq F^{m+1}Y$$

and therefore induces

$$\overline{L}_m : F^m X / F^{m+1} X \longrightarrow F^m Y / F^{m+1} Y.$$

At depth m, only the associated graded correction is visible. The cokernel below has finite length whenever the target graded piece has finite length, even if the completed problem is infinite. Finite cardinality requires additional finiteness of the underlying graded pieces. Its vanishing answers one precise question: whether the current approximation can be improved by one level.

Definition 23.1 (Depth obstruction)**.** The depth m obstruction of x_m is

$$\mathrm{Ob}_m(x_m) = [\Phi(x_m)] \in \operatorname{coker} \overline{L}_m,$$

where $\Phi(x_m)$ is first viewed in $F^m Y / F^{m+1} Y$.

Theorem 23.2 (One step lifting criterion)**.** *The approximation x_m has a correction $h_m \in F^m X$ such that*

$$\Phi(x_m + h_m) \in F^{m+1} Y$$

if and only if

$$\mathrm{Ob}_m(x_m) = 0.$$

When a correction exists, the set of its classes in

$$F^m X / F^{m+1} X$$

is a coset of $\operatorname{Ker} \overline{L}_m$*. If the graded pieces are vector spaces, this coset is an affine space.*

Proof. The next depth condition is equivalent to

$$\overline{L}_m[h_m] = -[\Phi(x_m)]$$

in $F^m Y / F^{m+1} Y$. This is solvable exactly when the displayed defect has zero cokernel class. If h_m and h'_m are two solutions, their difference lies in $\operatorname{Ker} \overline{L}_m$. Adding an element of the kernel to one solution gives another, so the solution set is a coset of the kernel. □

Corollary 23.3 (Finite branching count)**.** *Suppose the graded pieces are finite vector spaces over a finite field. If the depth obstruction vanishes, then the number of correction classes at depth m is*

$$|\operatorname{Ker} \overline{L}_m|.$$

If the obstruction does not vanish, the number is zero.

Proof. A nonempty affine space over a finite vector space has the same number of elements as that vector space. □

Lemma 23.4 (Well defined lift classes)**.** *Under the filtration congruence assumption, the condition*

$$\Phi(x) \in F^m Y$$

depends only on the class of x in $X/F^m X$. Fix a representative x_m of a depth m class. The map

$$[h] \longmapsto [x_m + h]_{m+1}$$

identifies correction classes $[h] \in F^m X/F^{m+1}X$ with the classes at depth $m+1$ that reduce to $[x_m]_m$. Under this identification, the solutions of the associated graded correction equation in theorem 23.2 are exactly the children whose defect lies in $F^{m+1}Y$. A different representative of the same parent gives a translated parametrization of the same set of children.

Proof. If $x' - x \in F^m X$, then filtration congruence preservation gives

$$\Phi(x') - \Phi(x) \in F^m Y.$$

Thus one defect lies in $F^m Y$ exactly when the other does. If $h - h' \in F^{m+1}X$, then $x_m + h$ and $x_m + h'$ define the same depth $m+1$ class. Conversely, two depth $m+1$ classes reducing to $[x_m]_m$ differ from x_m by elements of $F^m X$, and they are equal precisely when those elements agree modulo $F^{m+1}X$. The first order congruence shows that such a child has defect in $F^{m+1}Y$ exactly when its correction class solves the associated graded equation. Replacing x_m by another representative translates the correction coordinate by an element of $F^m X/F^{m+1}X$ and leaves the child classes themselves unchanged. □

Definition 23.5 (Lift tree)**.** Fix an initial approximation. A vertex at depth m is a class $[x]_m \in X/F^m X$ with $\Phi(x) \in F^m Y$ that reduces to the fixed initial class. A depth $m+1$ vertex is joined to its reduction in $X/F^m X$. By lemma 23.4, this rooted graph is independent of the chosen representatives. It is called the lift tree.

Theorem 23.6 (Finitely branching lift tree)**.** *Assume every level of the lift tree is nonempty and every vertex has only finitely many children. Then the tree contains an infinite branch. If X and Y are separated and complete and Φ is continuous, that branch determines an exact solution of $\Phi(x) = 0$.*

Proof. The first assertion is König's infinity argument. The root has finitely many children. Since vertices occur at arbitrarily large depths, at least one child has descendants at arbitrarily large depths. Choose such a child and repeat the argument at every chosen vertex. This gives one compatible vertex at each depth.

Choose representatives recursively. Once x_m represents the selected depth m vertex, choose a representative x_{m+1} of its selected child with

$$x_{m+1} - x_m \in F^m X.$$

The resulting sequence is Cauchy for the filtration topology. Each representative satisfies $\Phi(x_m) \in F^m Y$. The conclusion now follows from theorem 23.8. □

Corollary 23.7 (Unique lift branch)**.** *Suppose an initial approximation is fixed, every depth obstruction vanishes along its descendants, and every map $\overline{L}_m$ has zero kernel. Then at most one compatible lift branch extends the initial approximation.*

Proof. By theorem 23.2, the correction classes at each depth form a coset of $\operatorname{Ker}\overline{L}_m$. A zero kernel leaves at most one correction class, so induction gives at most one branch. □

Theorem 23.8 (Complete filtered lifting)**.** *Assume there is a compatible sequence*

$$x_{m+1} \equiv x_m \pmod{F^m X}$$

with

$$\Phi(x_m) \in F^m Y$$

for every m. Then x_m converges to an element $x_\infty \in X$ and

$$\Phi(x_\infty) = 0.$$

In particular, if the obstruction vanishes at every depth and compatible corrections are chosen, an exact solution exists.

Proof. For every fixed r, the congruences imply

$$x_n - x_m \in F^r X$$

whenever $n \geq m \geq r$. Thus (x_m) is Cauchy for the filtration topology. Completeness gives a limit x_∞.

Fix r. For every $m \geq r$, one has $\Phi(x_m) \in F^m Y \subseteq F^r Y$. The submodule $F^r Y$ is open and therefore closed in the filtration topology. Continuity of Φ gives $\Phi(x_m) \to \Phi(x_\infty)$, so $\Phi(x_\infty) \in F^r Y$. Since this holds for every r and Y is separated,

$$\Phi(x_\infty) \in \bigcap_r F^r Y = 0.$$

□

Theorem 23.9 (Naturality of filtered obstructions)**.** *Suppose a morphism of filtered lifting problems sends x_m to x'_m, intertwines L_m with L'_m, and sends $\Phi(x_m)$ to $\Phi'(x'_m)$ on the associated graded pieces. Then the induced cokernel map sends*

$$\operatorname{Ob}_m(x_m) \quad \textit{to} \quad \operatorname{Ob}'_m(x'_m).$$

Proof. Apply theorem 21.3 to the induced commuting square on the depth m graded pieces. □

Example 23.10 (Three lifting outcomes)**.** The scalar equation $x^2 - 2 = 0$ at $x = 3$ modulo 7 has invertible derivative. It has one correction modulo 49 and gives $x = 10$ modulo 49.

For $x^2 - 3 = 0$ at $x = 0$ modulo 3, the derivative is zero and the defect class is nonzero. No correction exists.

For $x^2 = 0$ at $x = 0$ modulo 3, the derivative and defect both vanish. Every correction digit lies in the kernel, so the lift branches.

Proposition 23.11 (Maurer Cartan lifting obstruction)**.** *Let $\mathcal{L}$ be a complete filtered differential graded Lie algebra over a ring in which two is invertible. Assume*

$$dF^i\mathcal{L} \subseteq F^i\mathcal{L}, \qquad [F^i\mathcal{L}, F^j\mathcal{L}] \subseteq F^{i+j}\mathcal{L}.$$

For a degree one element η, put

$$\Omega(\eta) = d\eta + \frac{1}{2}[\eta, \eta].$$

Suppose $\Omega(\eta) \in F^m\mathcal{L}^2$ with $m \geq 1$. For every $h \in F^m\mathcal{L}^1$,

$$\Omega(\eta + h) \equiv \Omega(\eta) + d_\eta h \pmod{F^{m+1}\mathcal{L}^2}, \qquad d_\eta = d + [\eta, \cdot].$$

The obstruction to improving η by one depth is therefore the class of $\Omega(\eta)$ in the cokernel of

$$\mathrm{gr}^m(d_\eta) : \mathrm{gr}^m \mathcal{L}^1 \longrightarrow \mathrm{gr}^m \mathcal{L}^2.$$

Proof. Direct expansion gives

$$\Omega(\eta + h) = \Omega(\eta) + dh + [\eta, h] + \frac{1}{2}[h, h].$$

The filtration rule places $[h, h]$ in $F^{2m}\mathcal{L}^2$, which is contained in $F^{m+1}\mathcal{L}^2$ because $m \geq 1$. The congruence follows. The last statement is theorem 23.2 applied to the induced graded correction map. □

24 Filtered obstruction complexes

A direct packet decomposition and a filtration behave differently. In a direct sum of subcomplexes, cohomology splits packet by packet. In a filtration, a class visible at one graded depth can be killed by a later differential.

There are two different mechanisms for a coarse obstruction to vanish. The first is packetwise vanishing. Every fine class is already zero. The second is cancellation through compatibility. Fine classes are nonzero, but later differentials pair them and remove them before the total cohomology is reached. A direct sum decomposition allows only the first mechanism. A filtration can allow both.

Standard Fact 24.1 (Spectral sequence of a finite filtered complex)**.** Let $C^\bullet$ be a cochain complex with a finite decreasing filtration by subcomplexes

$$C^\bullet = F^0C^\bullet \supseteq F^1C^\bullet \supseteq \cdots \supseteq F^{N+1}C^\bullet = 0.$$

There is a natural spectral sequence with

$$E_0^{p,q} = \mathrm{gr}^p C^{p+q}, \qquad E_1^{p,q} = H^{p+q}(\mathrm{gr}^p C^\bullet),$$

which converges to the associated graded object of $H^{p+q}(C^\bullet)$ with its induced filtration. A filtered cochain map induces a morphism of spectral sequences.

This is the standard filtered complex construction. We use it in the form given in [18, 12]. When $C^\bullet$ is a correction or deformation complex, the groups on its pages can be read as graded

obstruction candidates. For an arbitrary filtered complex they are simply cohomology classes of the associated graded problem.

Proposition 24.2 (Fine obstruction survival)**.** *A class on the first page contributes to the total obstruction only if it is represented by a permanent cycle, so it is neither killed nor identified as a boundary on any later page. If the spectral sequence degenerates at the first page, then $H^n(C^\bullet) = 0$ exactly when $E_1^{p,n-p} = 0$ for every p.*

Proof. Convergence identifies the last page with the associated graded object of the total cohomology. A first page class contributes to that object exactly when it determines a nonzero class on the last page. If the sequence degenerates at the first page, the associated graded pieces of $H^n(C^\bullet)$ are the groups $E_1^{p,n-p}$. A finite filtered module is zero exactly when all its associated graded pieces are zero. □

Corollary 24.3 (Direct packet complexes)**.** *Suppose*

$$C^\bullet = \bigoplus_{\alpha \in I} C_\alpha^\bullet$$

is a finite direct sum of subcomplexes. Then

$$H^n(C^\bullet) \cong \bigoplus_{\alpha \in I} H^n(C_\alpha^\bullet).$$

A total cohomology class vanishes if and only if every packet class vanishes.

Proof. The differential acts independently on the summands. Kernels and images therefore split as finite direct sums, and so does their quotient. □

Example 24.4 (A fine class killed at the next page)**.** Let $C^0 = \Bbbk x$ and $C^1 = \Bbbk y$, with differential $d(x) = y$. Give C^0 the filtration

$$F^0C^0 = \Bbbk x, \qquad F^1C^0 = 0,$$

and give C^1 the filtration

$$F^0C^1 = F^1C^1 = \Bbbk y, \qquad F^2C^1 = 0.$$

The associated graded differential on the first page is zero in the relevant positions, so both x and y appear as nonzero fine classes. The next differential sends the class of x to the class of y. The total complex is acyclic. Thus nonzero fine obstruction classes can cancel through a later refinement differential.

Remark 24.5. The spectral sequence does not guarantee that a total obstruction vanishes. It records which fine classes survive and which are removed by compatibility across depths.

25 Nonlinear elimination

Let U, V, Y_P, Y_Q be Banach spaces. On an open neighborhood of $(u_0, v_0) \in U \times V$, split the variables and equations as

$$x = (u, v), \qquad F(u, v) = \begin{pmatrix} F_P(u, v) \\ F_Q(u, v) \end{pmatrix}.$$

Assume F takes values in $Y_P \times Y_Q$, is twice continuously Fréchet differentiable near (u_0, v_0), satisfies $F_Q(u_0, v_0) = 0$, and

$$A = D_v F_Q(u_0, v_0)$$

is a Banach space isomorphism from V to Y_Q. The implicit function theorem gives a local map $v = \varphi(u)$, with $\varphi(u_0) = v_0$, satisfying

$$F_Q(u, \varphi(u)) = 0.$$

Define the reduced equation

$$R(u) = F_P(u, \varphi(u)).$$

Theorem 25.1 (Exact local elimination)**.** *After the neighborhoods are chosen small enough, the equation*

$$F(u, v) = 0$$

is equivalent to

$$v = \varphi(u) \qquad \text{and} \qquad R(u) = 0.$$

Thus eliminating the hidden variable loses no local solutions.

Proof. The hidden equation $F_Q(u, v) = 0$ has the unique local solution $v = \varphi(u)$ by the implicit function theorem. After this identity is imposed, the remaining component of F is exactly $R(u)$. □

Theorem 25.2 (Nonlinear Schur derivative)**.** *At* (u_0, v_0)*,*

$$DR = D_u F_P - D_v F_P A^{-1} D_u F_Q.$$

Proof. Differentiate $F_Q(u, \varphi(u)) = 0$:

$$D_u F_Q + A D\varphi = 0.$$

Thus

$$D\varphi = -A^{-1} D_u F_Q.$$

Substitute this into

$$DR = D_u F_P + D_v F_P D\varphi.$$

□

The second derivative records the first nonlinear feedback from the hidden variables.

Theorem 25.3 (Second Schur derivative)**.** *For retained directions* h, k*, set*

$$\widehat{h} = (h, D\varphi h), \qquad \widehat{k} = (k, D\varphi k).$$

Then

$$D^2\varphi[h,k] = -A^{-1}D^2F_Q[\widehat{h},\widehat{k}]$$

and

$$D^2R[h,k] = D^2F_P[\widehat{h},\widehat{k}] - D_vF_PA^{-1}D^2F_Q[\widehat{h},\widehat{k}].$$

All derivatives on the right are evaluated at (u_0, v_0).

Proof. Differentiate the identity $F_Q(u,\varphi(u)) = 0$ twice. The chain rule gives

$$D^2F_Q[\widehat{h},\widehat{k}] + AD^2\varphi[h,k] = 0.$$

This proves the first formula. Differentiate $R(u) = F_P(u,\varphi(u))$ twice and substitute the first formula. □

Corollary 25.4 (Packet jets of the reduced equation)**.** *If the retained and hidden spaces carry packet systems, then the first and second packet jets of* R *are obtained by projecting the formulas in theorems 25.2 and 25.3. Hidden packet feedback is carried by the factors involving* A^{-1}.

Proof. Apply the packet projectors to the input and output slots of the two derivative formulas. □

Example 25.5 (Two retained variables and one hidden variable)**.** Suppose the derivative of F at a solution has block form

$$DF = \begin{pmatrix} A & B \\ C & D \end{pmatrix},$$

where D is invertible. Eliminating the hidden variable replaces the retained derivative A by

$$A - BD^{-1}C.$$

The feedback term is exact. Keeping only A would ignore the path from retained variables into the hidden variable and back.

Remark 25.6. Elimination and obstruction reduction address different situations. The Schur formula assumes that the hidden derivative is invertible. Lyapunov Schmidt reduction allows a nontrivial kernel and cokernel, then leaves a reduced obstruction equation on the kernel directions.

26 Finite transport and holonomy

Let $\Gamma = (V, E)$ be a finite connected graph and fix a vector space X. At every vertex take one labeled copy of X. Every oriented edge $u \to v$ carries an invertible transport

$$T_{vu} \in \mathrm{GL}(X), \qquad T_{uv} = T_{vu}^{-1}.$$

Equivalently, one may begin with separate but isomorphic vertex fibers and choose one frame in each fiber. The matrices below are then the transports in those frames. For a path $\gamma = (v_0, \dots, v_r)$, set

$$T_\gamma = T_{v_rv_{r-1}} \cdots T_{v_1v_0}.$$

For a loop, this product is its holonomy. In the based loop group, $\gamma_2\gamma_1$ means first traverse γ_1 and then γ_2, so $T_{\gamma_2\gamma_1} = T_{\gamma_2}T_{\gamma_1}$. Standard graph terminology may be found in [5].

Local solvability appears here as the existence of every edge map. Global solvability asks whether all local frames can be chosen at once. A spanning tree has no cycles, so its edge data can always be removed by a gauge. Every remaining edge closes a loop and carries the full global obstruction for that loop.

Definition 26.1 (Gauge change)**.** A gauge is a family $G_v \in \mathrm{GL}(X)$. It changes the edge transports to

$$T'_{vu} = G_v T_{vu} G_u^{-1}.$$

Proposition 26.2 (Gauge covariance)**.** *For a loop γ based at v_0,*

$$\mathrm{Hol}'(\gamma) = G_{v_0} \mathrm{Hol}(\gamma) G_{v_0}^{-1}.$$

Thus the conjugacy class of each loop holonomy is gauge invariant.

Proof. Every intermediate gauge factor cancels with its inverse. The initial and terminal factors remain at the base vertex. □

Theorem 26.3 (Spanning tree normalization)**.** *Choose a root and a spanning tree $\mathcal{T}$ of Γ. There is a gauge in which every tree edge transport is the identity. In that gauge, the transport on each edge outside the tree is the holonomy of its fundamental cycle, with the orientation used to define that cycle.*

Proof. Set the gauge at the root to the identity. Propagate gauges along the unique tree paths so that every tree edge becomes the identity. An edge outside the tree closes one fundamental cycle. Since every tree factor is now the identity, the remaining edge transport is the cycle product. □

Theorem 26.4 (Cycle gluing criterion)**.** *There are vertex coordinates in which every edge transport is the identity if and only if every fundamental cycle holonomy is the identity.*

Proof. If all edge transports are identity maps, every loop product is the identity. Conversely, use theorem 26.3. Identity fundamental holonomy forces every edge outside the tree to become the identity as well. □

Corollary 26.5 (Number of independent cycle tests)**.** *For a connected graph with $|V|$ vertices and $|E|$ unoriented edges, a spanning tree leaves*

$$|E| - |V| + 1$$

fundamental cycle holonomies.

Proof. A spanning tree has $|V| - 1$ edges. Every remaining edge creates one fundamental cycle. □

26.1 Intertwining two transport systems

Let A be a transport system on copies of a fixed vector space V_0, and let B be a transport system on copies of a fixed vector space W_0. Fix a root r and an isomorphism

$$U_r : V_0 \longrightarrow W_0.$$

Theorem 26.6 (Fundamental holonomy intertwining criterion)**.** *The following conditions are equivalent.*

1. *There is a family of isomorphisms $U_v : V_0 \to W_0$ with the prescribed root value U_r and satisfying*

$$U_v A_{vu} = B_{vu} U_u$$

   for every oriented edge $u \to v$.

2. *For every fundamental loop γ based at r,*

$$U_r \operatorname{Hol}_A(\gamma) = \operatorname{Hol}_B(\gamma) U_r.$$

3. *The same identity holds for every loop based at r.*

When these conditions hold, the family (U_v) is uniquely determined by U_r.

Proof. If the edge intertwiners exist, multiply their equations around a loop. Every intermediate vertex map cancels. This proves that the first condition implies the third, and the third implies the second.

Conversely, choose a spanning tree. Starting from U_r, propagate U_v along the unique tree path by the rule

$$U_v = B_{vu} U_u A_{vu}^{-1}$$

on every tree edge. This is the only possible extension satisfying the tree edge equations. An edge outside the tree closes one fundamental loop. Its intertwining equation holds exactly when U_r intertwines the two corresponding fundamental holonomies. The second condition therefore supplies every remaining edge equation. Uniqueness follows from propagation along the tree. □

Corollary 26.7 (Scalar holonomy bridge)**.** *Suppose every vertex space is one dimensional. Two transport systems are gauge equivalent if and only if their holonomies agree on a fundamental cycle basis.*

Proof. Choose a nonzero root isomorphism. Scalar loop maps commute with it, so theorem 26.6 reduces to equality of the cycle scalars. □

Theorem 26.8 (Gauge classification of transports)**.** *Let $G \subseteq \mathrm{GL}(X)$ be the chosen structure group and fix a root r. Restrict both edge transports and gauge values to G. Gauge equivalence classes of G valued transport systems are naturally identified with*

$$\operatorname{Hom}(\pi_1(\Gamma, r), G)/G,$$

where G acts by conjugation. After a spanning tree and orientations of the remaining edges are chosen, this set is identified with simultaneous conjugacy classes of

$$b_1 = |E| - |V| + 1$$

tuples in G.

Proof. A transport system assigns a group element to every oriented edge, with inversion under reversal. The product along a based loop gives a homomorphism from the free group $\pi_1(\Gamma, r)$ to

G. A gauge change conjugates this homomorphism by the gauge value at the root. Conversely, normalize a transport system on a spanning tree. The remaining edge values prescribe the images of a free basis of $\pi_1(\Gamma, r)$, and every homomorphism arises in this way. A further gauge preserving the tree normalization is constant at all vertices and acts by simultaneous conjugation. The number of non tree edges is b_1. □

Corollary 26.9 (Abelian cohomology class)**.** *When the structure group A is abelian, gauge classes of transports are naturally*

$$H^1(\Gamma, A) \cong \mathrm{Hom}(H_1(\Gamma, \mathbb{Z}), A).$$

After a basis of $H_1(\Gamma, \mathbb{Z})$ is chosen, this group is identified with A^{b_1}.

Proof. An edge assignment is an A valued one cochain. A gauge change multiplies it by a coboundary. A graph has no two cells, so every one cochain is a cocycle. The usual computation of the first cohomology of a connected graph gives A^{b_1}. □

Corollary 26.10 (Projective transport)**.** *The same normalization and gluing statements hold for transports valued in a projective linear group. A projective transport is globally trivial exactly when all fundamental holonomies are identity elements in that projective group.*

Proof. Apply theorems 26.3 and 26.4 with the projective linear group as the structure group. □

Example 26.11 (A square loop)**.** A square graph has four vertices and four edges, so it has one fundamental cycle. After a tree gauge, three edges are identities and the fourth is the full loop holonomy. The system glues to one global frame exactly when that remaining matrix is the identity.

27 Smooth projector transport

Finite graph transport has a continuous counterpart. Let H be a Hilbert space and let

$$\mathcal{P}(t) = (P_\alpha(t))_{\alpha \in I}$$

be a finite family of complete orthogonal packet systems that is continuously differentiable in operator norm for $t \in [0, T]$, where $T \geq 0$. Here every $P_\alpha(t)$ is a bounded self-adjoint projection, so $P_\alpha(t)^* = P_\alpha(t) = P_\alpha(t)^2$. Zero projections are permitted in this section and contribute zero to all generator sums. When $T = 0$, every partition product is the empty product I. Thus

$$P_\alpha(t)P_\beta(t) = 0 \quad \text{for } \alpha \neq \beta, \qquad \sum_\alpha P_\alpha(t) = I.$$

Define the canonical generator

$$K(t) = \sum_\alpha \dot{P}_\alpha(t) P_\alpha(t).$$

Lemma 27.1 (Complete packet commutator identity)**.** *For every packet label β,*

$$\dot{P}_\beta(t) = [K(t), P_\beta(t)].$$

The generator is also skew adjoint:

$$K(t)^* = -K(t).$$

Proof. Differentiating $P_\alpha^2 = P_\alpha$ gives

$$P_\alpha \dot{P}_\alpha P_\alpha = 0.$$

Differentiating $P_\beta P_\alpha = 0$ for $\alpha \neq \beta$ gives

$$P_\beta \dot{P}_\alpha = -\dot{P}_\beta P_\alpha.$$

Hence

$$K P_\beta = \dot{P}_\beta P_\beta$$

and

$$P_\beta K = -\dot{P}_\beta (I - P_\beta).$$

Their difference is $\dot{P}_\beta$. Also,

$$K + K^* = \sum_\alpha (\dot{P}_\alpha P_\alpha + P_\alpha \dot{P}_\alpha) = \sum_\alpha \dot{P}_\alpha = 0.$$

□

Theorem 27.2 (Canonical packet transport)**.** *Let $U(t)$ solve*

$$\dot{U}(t) = K(t)U(t), \qquad U(0) = I.$$

Then $U(t)$ is unitary and transports every packet exactly:

$$P_\alpha(t)U(t) = U(t)P_\alpha(0)$$

for all α.

Proof. The continuous bounded generator has an evolution inverse. Indeed, solve $\dot{V} = -VK$ with $V(0) = I$. Differentiation gives $(VU)' = 0$, hence $VU = I$. The product $W = UV$ satisfies $\dot{W} = KW - WK$ and $W(0) = I$; uniqueness of this bounded linear equation gives $UV = I$. Thus U is invertible.

Skew adjointness of $K(t)$ gives

$$\frac{d}{dt}(U(t)^* U(t)) = 0,$$

so $U(t)$ is unitary. By lemma 27.1,

$$\frac{d}{dt}(U(t)^{-1} P_\alpha(t) U(t)) = U(t)^{-1}(\dot{P}_\alpha - [K, P_\alpha])U(t) = 0.$$

The expression equals $P_\alpha(0)$ at time zero. □

Corollary 27.3 (One moving subspace)**.** *For one orthogonal projection $P(t)$ and its complement, the canonical generator is*

$$K(t) = [\dot{P}(t), P(t)].$$

The transport in theorem 27.2 maps $\operatorname{im} P(0)$ *unitarily onto* $\operatorname{im} P(t)$.

Proof. Apply the complete packet formula to $P(t)$ and $I - P(t)$, then simplify the sum. □

Definition 27.4 (Direct rotation of nearby packet systems)**.** Let $(P_\alpha)_{\alpha\in I}$ and $(Q_\alpha)_{\alpha\in I}$ be complete orthogonal packet systems on the same Hilbert space. Put

$$S(\mathcal{Q},\mathcal{P})=\sum_{\alpha\in I}Q_\alpha P_\alpha.$$

When $S(\mathcal{Q},\mathcal{P})$ is invertible, define

$$\mathcal{U}(\mathcal{Q},\mathcal{P})=S(\mathcal{Q},\mathcal{P})\big(S(\mathcal{Q},\mathcal{P})^*S(\mathcal{Q},\mathcal{P})\big)^{-1/2}.$$

Proposition 27.5 (Exact direct packet rotation)**.** *If*

$$\max(\{0\}\cup\{\|P_\alpha-Q_\alpha\|:\alpha\in I\})<1,$$

then $S(\mathcal{Q},\mathcal{P})$ is invertible. The operator $\mathcal{U}(\mathcal{Q},\mathcal{P})$ is unitary and satisfies

$$Q_\alpha\mathcal{U}(\mathcal{Q},\mathcal{P})=\mathcal{U}(\mathcal{Q},\mathcal{P})P_\alpha$$

for every α.

Proof. The identities

$$Q_\alpha S(\mathcal{Q},\mathcal{P})=S(\mathcal{Q},\mathcal{P})P_\alpha=Q_\alpha P_\alpha$$

follow from pairwise orthogonality. Also

$$S^*S=\sum_\alpha P_\alpha Q_\alpha P_\alpha,\qquad SS^*=\sum_\alpha Q_\alpha P_\alpha Q_\alpha.$$

Put $\delta=\max(\{0\}\cup\{\|P_\alpha-Q_\alpha\|:\alpha\in I\})$. On $P_\alpha H$,

$$P_\alpha Q_\alpha P_\alpha=P_\alpha-P_\alpha(P_\alpha-Q_\alpha)P_\alpha,$$

so this block is bounded below by $1-\delta>0$. The same argument on $Q_\alpha H$ applies to $Q_\alpha P_\alpha Q_\alpha$. Thus both S^*S and SS^* are invertible. It follows that S is injective and surjective. The polar factor $\mathcal{U}=S(S^*S)^{-1/2}$ is unitary. Since S^*S commutes with every P_α, the intertwining identity for S passes to $\mathcal{U}$. □

Lemma 27.6 (One step direct rotation estimate)**.** *Assume $\mathcal{P}(t)$ is twice continuously differentiable in operator norm on $[0,T]$. There are constants $h_0>0$ and $C>0$ such that, whenever $t,t+h\in[0,T]$ and $|h|\le h_0$,*

$$\|\mathcal{U}(\mathcal{P}(t+h),\mathcal{P}(t))-I-hK(t)\|\le Ch^2.$$

The constants can be chosen uniformly in t.

Proof. Taylor's theorem in the Banach space of bounded operators gives

$$P_\alpha(t+h)=P_\alpha(t)+h\dot P_\alpha(t)+E_\alpha(t,h),\qquad\|E_\alpha(t,h)\|\le C_1h^2,$$

uniformly in t and α. Hence

$$S(\mathcal{P}(t+h),\mathcal{P}(t))=I+hK(t)+E(t,h),\qquad\|E(t,h)\|\le C_2h^2.$$

Since $K(t)^* = -K(t)$,

$$S^*S = I + E_2(t,h), \qquad \|E_2(t,h)\| \le C_3 h^2.$$

For small h, the spectrum of S^*S lies in a fixed compact subinterval of $(0,\infty)$. The functional calculus for $z^{-1/2}$ gives

$$(S^*S)^{-1/2} = I + E_3(t,h), \qquad \|E_3(t,h)\| \le C_4 h^2.$$

Multiplying the last two expansions proves the estimate. □

Theorem 27.7 (Direct rotations converge to packet transport)**.** *Assume the complete packet family $\mathcal{P}(t)$ is twice continuously differentiable in operator norm on $[0,T]$. For a partition*

$$0 = t_0 < t_1 < \cdots < t_N = T$$

with sufficiently small mesh, define

$$V_\pi = \mathcal{U}(\mathcal{P}(t_N), \mathcal{P}(t_{N-1})) \cdots \mathcal{U}(\mathcal{P}(t_1), \mathcal{P}(t_0)).$$

Then

$$V_\pi \longrightarrow U(T)$$

in operator norm, where U is the canonical transport from theorem 27.2. On a fixed compact interval the error is bounded by a constant times the mesh.

Proof. Let $U(t,s)$ be the exact propagator generated by K. Since K is continuously differentiable, the integral equation for $U(t+h,t)$ gives a uniform estimate

$$\|U(t+h,t) - I - hK(t)\| \le C_1 h^2.$$

By lemma 27.6, the direct rotation V_j from t_j to t_{j+1} therefore satisfies

$$\|V_j - U(t_{j+1},t_j)\| \le C_2 h_j^2, \qquad h_j = t_{j+1} - t_j.$$

Both V_j and the exact propagator are unitary. A telescoping product comparison gives

$$\|V_\pi - U(T,0)\| \le \sum_j \|V_j - U(t_{j+1},t_j)\| \le C_2 \sum_j h_j^2 \le C_2 T|\pi|.$$

This proves norm convergence and the stated first order global bound. □

Theorem 27.8 (Convergence of sampled transports)**.** *Assume K is Lipschitz on $[0,T]$. For a partition*

$$0 = t_0 < t_1 < \cdots < t_N = T,$$

define

$$U_\pi = \exp\big((t_N - t_{N-1})K(t_{N-1})\big) \cdots \exp\big((t_1 - t_0)K(t_0)\big).$$

As the mesh $|\pi|$ tends to zero,

$$U_\pi \longrightarrow U(T)$$

in operator norm. There is a constant C, depending only on T, the supremum norm of K, and its Lipschitz constant, such that

$$\|U_\pi - U(T)\| \le C|\pi|.$$

Proof. On each interval, compare the exact propagator with the frozen exponential. Variation of constants and the Lipschitz bound give a local error bounded by a constant times the square of the interval length. Every factor is unitary, so multiplication does not amplify norms. Summing the local errors gives a bound by a constant times

$$\sum_j (t_{j+1} - t_j)^2 \le T|\pi|.$$

□

Corollary 27.9 (Continuous holonomy from graph products)**.** *Assume the complete packet family is twice continuously differentiable in operator norm on $[0,T]$. If $P_\alpha(0) = P_\alpha(T)$ for every α, then $U(T)$ preserves every initial packet. Its restriction to each packet is the holonomy of the closed packet path. The products in theorems 27.7 and 27.8 converge to these holonomies. The direct rotation product has the added property that every factor maps each sampled packet exactly to its successor.*

Proof. The transport identity in theorem 27.2 gives packet preservation at the endpoint. The two convergence statements follow from theorems 27.7 and 27.8. □

Example 27.10 (A rotating line)**.** In $\mathbb{R}^2$, let

$$v(t) = (\cos t, \sin t)$$

and let $P(t)$ be the orthogonal projection onto the line spanned by $v(t)$. The canonical transport is the ordinary rotation matrix

$$U(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

It carries the initial line and its orthogonal complement to the two moving packets. Products of short rotations converge to the same transport.

Standard Fact 27.11 (Curvature of a projected connection)**.** Let P be a smooth orthogonal projector valued map on a parameter manifold. The connection

$$\nabla = P\,d$$

on the range bundle has curvature

$$\mathcal{F} = P(dP \wedge dP)P.$$

Proof. For a section s satisfying $Ps = s$,

$$\nabla^2 s = P\,d(P\,ds) = P(dP) \wedge ds.$$

Since $ds = (dP)s + P\,ds$ and $P(dP)P = 0$, the surviving term is

$$P(dP \wedge dP)Ps.$$

□

28 Realization data and limits of the calculus

The continuous theorem transports a declared projector family. It does not construct the family, prove a continuum limit for a physical model, or show that its holonomy is trivial. The curvature formula in standard fact 27.11 is classical. Relating it to global holonomy still requires geometric information as in [7, 8, 10].

To apply the results in another setting, one must identify the following objects:

1. an ambient nonlinear equation or continuation problem
2. the allowed correction space
3. a linear correction map or a correction complex
4. the defect whose class is to be tested
5. a packet decomposition or filtration compatible with the correction data
6. transition maps between refinement levels
7. transport maps when local solutions must be glued.

Once these objects are fixed, the cokernel and cone constructions are functorial by theorems 21.3 and 21.11. Packet splitting and descent follow from theorem 21.4, proposition 21.6, and corollary 21.12. Relative loss under coarsening is measured by theorem 21.7. Filtered lifting follows from theorems 23.2, 23.6 and 23.8. Gluing and comparison of transport systems follow from theorems 26.4 and 26.6. Smooth packet transport follows from theorems 27.2 and 27.7.

Several limits remain explicit. Exact packet coordinates are a representation theorem, not a general solution theorem. The cokernel is canonical relative to the correction map, but the correction map itself belongs to the problem data. The cokernel may be infinite dimensional. A split Hilbert projector requires closed range. Coarse vanishing can erase a nonzero fine obstruction. A vanishing linear obstruction does not remove higher nonlinear terms. A spectral sequence records possible cancellation, but it does not force cancellation. Finite graph holonomy and smooth projector transport do not establish a continuum field theory.

These results do not assert that all obstructions vanish. Nonlinear continuation failures can be placed in exact quotient or cohomology objects whenever a correction problem is specified. Those objects admit packet decomposition, filtered refinement, natural transport, and holonomy tests under explicit hypotheses.

Use of language model tools

The author used an OpenAI language model to help reorganize and draft parts of this article. The model suggested formulations and proof approaches for the polynomial tree theorem in section 18, the cokernel, relative obstruction, and mapping cone results in section 21, the Lyapunov Schmidt and filtered lifting arguments, the holonomy intertwining theorem in section 26, and the direct packet rotation theorem in section 27. It was also used during a later consistency audit to locate missing regularity and filtration hypotheses. The author checked the statements, proofs, and citations and accepts full responsibility for them. The model is not an author.

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